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The exocyst complex controls multiple events in the pathway of regulated exocytosis

Sofía Suárez Freire, Sebastián Pérez-Pandolfo, Sabrina M Fresco, Julián Valinoti, Eleonora Sorianello, Pablo Wappner, Mariana Melani

Fundación Instituto Leloir, Buenos Aires, Argentina • Consejo Nacional De Investigaciones Científicas Y Técnicas (CONICET), Buenos Aires, Argentina • Laboratorio de Regulación Hipofisaria. Instituto de Medicina y Biología Experimental (IBYME-CONICET), Buenos Aires, Argentina • Departamento De Fisiología, Biología Molecular Y Celular, Facultad De Ciencias Exactas Y Naturales, Universidad De Buenos Aires, Buenos Aires, Argentina

eLife Assessment

This study makes an **important** contribution by characterizing the role of the exocyst in secretory granule exocytosis in the Drosophila larval salivary gland. The results are **solid** and lead to the novel interpretation that the exocyst participates not only in exocytosis, but also in earlier steps of secretory granule biogenesis and maturation. However, the authors are urged to provide additional proof that the exocyst subunit knockdowns were effective and to acknowledge the possibility that inactivation of an essential exocytosis component could indirectly affect other parts of the secretory pathway.

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Abstract

Eukaryotic cells depend on exocytosis to direct intracellularly synthesized material towards the extracellular space or the plasma membrane, so exocytosis constitutes a basic function for cellular homeostasis and communication between cells. The secretory pathway includes biogenesis of secretory granules (SGs), their maturation and fusion with the plasma membrane (exocytosis), resulting in release of SG content to the extracellular space. The larval salivary gland of *Drosophila melanogaster* is an excellent model for studying exocytosis. This gland synthesizes mucins that are packaged in SGs that sprout from the *trans*-Golgi network and then undergo a maturation process that involves homotypic fusion, condensation and acidification. Finally, mature SGs are directed to the apical domain of the plasma membrane with which they fuse, releasing their content into the gland lumen. The exocyst is a hetero-octameric complex that participates in tethering of vesicles to the plasma membrane during constitutive exocytosis. By precise temperature-dependent gradual activation of the Gal4-UAS expression system, we have induced different levels of silencing of exocyst complex subunits, and identified three temporarily distinctive steps of the regulated exocytic pathway where the exocyst is critically required: SG biogenesis, SG maturation and

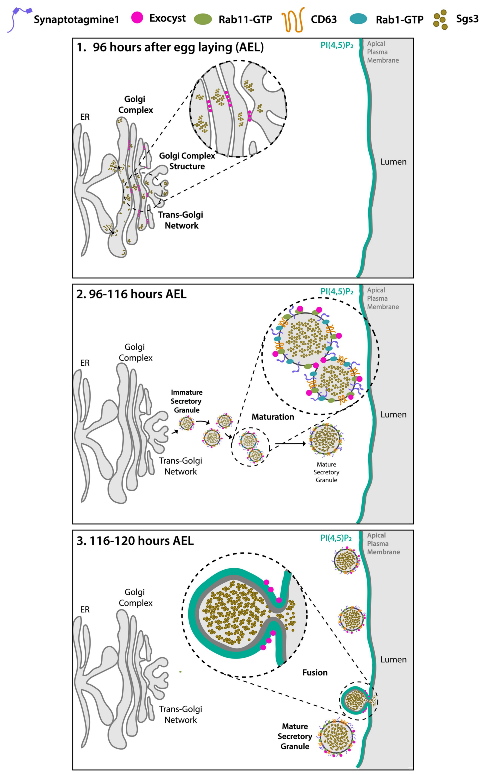
SG exocytosis. Our results shed light on previously unidentified functions of the exocyst along the exocytic pathway. We propose that the exocyst acts as a general tethering factor in various steps of this cellular process.

Introduction

Protein secretion is a fundamental process for communication between eukaryotic cells and therefore, for organismal homeostasis, reproduction and survival. Two types of secretory processes can be distinguished based on the rate and mode of regulation of secretory vesicle release: Constitutive secretion and regulated secretion (Morgan, 1995). In constitutive secretion, secretory vesicles are exocytosed as they are produced. This process takes place in all eukaryotic cells, mainly to maintain homeostasis of the plasma membrane and the extracellular matrix. In contrast, regulated secretion takes place in specialized cell types, such as endocrine and exocrine cells, as well as in cells of the immune system such as platelets and neutrophils (Aggarwal, Jennings, Manning, & Cameron, 2023; Ley et al., 2018). These cell types produce specialized secretory vesicles known as secretory granules (SGs), which are released in response to specific stimuli (Kogel & Gerdes, 2010). SGs sprout from the *trans*-Golgi network (TGN) as immature vesicles, incompetent for fusion with the plasma membrane. Maturation of SGs involves homotypic fusion and acquisition of membrane proteins, which are required for SG delivery and fusion with the plasma membrane. Finally, stimulus-driven fusion of SGs with the plasma membrane results in cargo release to the extracellular milieu (Omar-Hmeadi & Idevall-Hagren, 2021; Sugita, 2008).

The salivary gland of *Drosophila melanogaster* larvae provides several advantages for the identification of regulators of SG biogenesis, maturation and exocytosis (Burgess et al., 2012; de la Riva-Carrasco et al., 2021; Neuman & Bashirullah, 2018; Rousso, Schejter, & Shilo, 2016). At late third larval instar, salivary glands synthesize a series of mucins, collectively known as Glue proteins, which become packed in SGs. Approximately four hours before the onset of pupariation, concerted exocytosis of mucin-containing SGs, followed by extrusion of the mucins out of the prepupal body are essential for gluing the puparium to the substratum (Borne, Kovalev, Gorb, & Courtier-Orgogozo, 2020). Interestingly, these three events of SG development (biogenesis, maturation and exocytosis) occur sequentially and only once in each cell of the salivary gland, that will later on be degraded during metamorphosis (Duan et al., 2020; Tracy, Velentzas, & Baehrecke, 2016). The tagging of one of these mucins with fluorophores (Sgs3-GFP or Sgs3-dsRed), (Biyasheva, Do, Lu, Vaskova, & Andres, 2001) combined with the large size of salivary gland cells and their SGs, have allowed high resolution imaging and real time traceability of SG biogenesis, maturation and secretion, leading to the identification and characterization of dozens of factors required along the exocytic pathway (Burgess et al., 2011; Neuman et al., 2022; Reynolds, Zhang, Tran, & Ten Hagen, 2019; Torres, Rosa-Ferreira, & Munro, 2014; Tran, Masedunskas, Weigert, & Ten Hagen, 2015).

The exocyst is a hetero-octameric protein complex identified and initially characterized in the budding yeast, and later found to be conserved across all eukaryotic organisms (Hsu et al., 1996; Novick, Field, & Schekman, 1980; Novick et al., 1995; TerBush, Maurice, Roth, & Novick, 1996). Yeast cells bearing mutations in exocyst subunits display intracellular accumulation of secretory vesicles and defects in exocytosis (Govindan, Bowser, & Novick, 1995; TerBush et al., 1996). Molecularly, the exocyst complex participates in vesicle tethering to the plasma membrane prior to SNARE-mediated fusion (An et al., 2021). Exocyst complex malfunction has been therefore associated with tumor growth and invasion, as well as with development of ciliopathies, among other pathological conditions (Luo, Zhang, Luca, & Guo, 2013; Mavor et al., 2016; Thapa et al., 2012; Whyte & Munro, 2002; B. Wu & Guo, 2015).



Graphical abstract.

Proposed model of the action of the exocyst in maintenance of normal Golgi complex structure, maturation and exocytosis of secretory granules in *Drosophila* larval salivary gland cells.

1-Before secretory granule (SG) biogenesis (<96 hours AEL), the exocyst (pink dots) localizes at the Golgi complex, where it is required to maintain the normal Golgi structure. The mucine Sgs3 (brown dots) moves through the secretory pathway from the endoplasmic reticulum (ER) to the Golgi complex, from where immature SGs containing the mucine sprout out. 2-After sprouting, SGs undergo maturation (96-116 hours AEL). During maturation, the exocyst localizes in between immature SGs, where it is required for homotypic fusion. The exocyst is also required for incorporation of maturation factors to the membrane of SGs. These maturation factors include Syt-1 (purple line), DC63 (orange line), Rab11 (green oval) and Rab1 (light blue oval). At this stage, the exocyst no longer localizes at the Golgi complex. 3-When maturation has been completed SGs fuse with the apical plasma membrane and exocytosis takes place. During exocytosis (116-120 hours AEL), the exocyst localizes at mature SGs, in contact with the apical plasma membrane, where it is required for tethering and subsequent fusion, prior to release of the SG content to the salivary gland lumen.

In this work, we have utilized the *Drosophila* salivary gland to carry out a methodic analysis of the requirement of each of the eight subunits of the exocyst along the regulated exocytic pathway. By inducing temperature-dependent gradual downregulation of the expression of each of the subunits, we discovered novel functions of the exocyst complex in regulated exocytosis, namely in SG biogenesis, SG maturation and homotypic fusion, as well as in SG fusion with the plasma membrane. We propose that the exocyst complex participates in all these processes as a general tethering factor.

Results

Characterization of secretory granule progression during salivary gland development

Salivary glands of *Drosophila melanogaster* larvae fulfill different functions during larval development. Mostly, they act as exocrine glands producing non-digestive enzymes during the larval feeding period, as well as mucins when the larva is about to pupariate (Costantino et al., 2008; Farkas et al., 2014). More recently, salivary glands have been proposed to behave as endocrine organs as well, secreting a yet unidentified factor that regulates larval growth (Li et al., 2022). Biosynthesis of mucins produced by salivary glands, named Salivary gland secreted proteins (Sgs), begins at the second half of the third larval instar in response to an ecdysone peak (Biyasheva et al., 2001). After being glycosylated at the Endoplasmic Reticulum (ER) and the Golgi Complex (GC), mucins are packed in SGs. Following subsequent developmentally-controlled hormonal stimuli, SGs are massively exocytosed, releasing the mucins to the gland lumen, and finally expelling them outside of the puparium, and gluing it to the substratum (Biyasheva et al., 2001). We used Sgs3-GFP or Sgs3-dsRed transgenic lines to follow this process *in vivo*. In wandering larvae, Sgs3-dsRed can be detected in salivary glands (**Figure 1A**’), while in prepupae, Sgs3-dsRed has been secreted out from the puparium, and is no longer detectable in salivary glands (**Figure 1B**’). To have a better temporal and spatial resolution of this process, we dissected and analyzed by confocal microscopy salivary glands expressing Sgs3-GFP at 4 h intervals starting at 96 h after egg laying (AEL). Expression of Sgs3-GFP begins at 96-100 h AEL, and can be detected at the distal region of the gland (**Figure 1C**). Later on, Sgs3-GFP expression expands to more proximal cells, and at 116 h AEL, the mucin becomes detectable in the whole gland, with the exception of ductal cells which do not express Sgs3 (**Figure 1C**) (Biyasheva et al., 2001). Thereafter, at 116-120 h AEL, in response to an ecdysone peak, SGs fuse with the apical plasma membrane, and release their content to the gland lumen (**Figure 1C**).

Detailed observation of distal cells of salivary glands revealed that SGs enlarge from 96 h AEL onwards (**Figure 1D**), (Ma & Brill, 2021b; Neuman, Lee, Selegue, Cavanagh, & Bashirullah, 2021), a phenomenon that we quantified by measuring SG diameter (**Figure 1D** and **Supplementary Table 1**). At 96-100 h AEL nascent SGs are smaller than 1 µm in diameter. Later, and up to 112 h AEL, most SGs are smaller than 3 µm in diameter, and classified them as “Immature SGs”. From 112 h AEL onwards most SGs are bigger than 3 µm in diameter and classified as “Mature SGs” (**Figure 1D** and **Supplementary Table 1**). The maximal SG diameter that we detected was 7.13 µm at 116-120 AEL, just before exocytosis (**Supplementary Table 1**).

The eight subunits of the exocyst complex are required for regulated exocytosis of secretory granules

Fluorophore-tagged Sgs3 can also be used to screen for exocytosis mutants in which Sgs3-GFP is retained inside salivary glands at the prepupal stage (de la Riva-Carrasco et al., 2021; Ma et al., 2020). Using this approach, we found that the exocyst complex is apparently required for exocytosis of SGs, since silencing of any of the eight subunits of the complex results in retention of

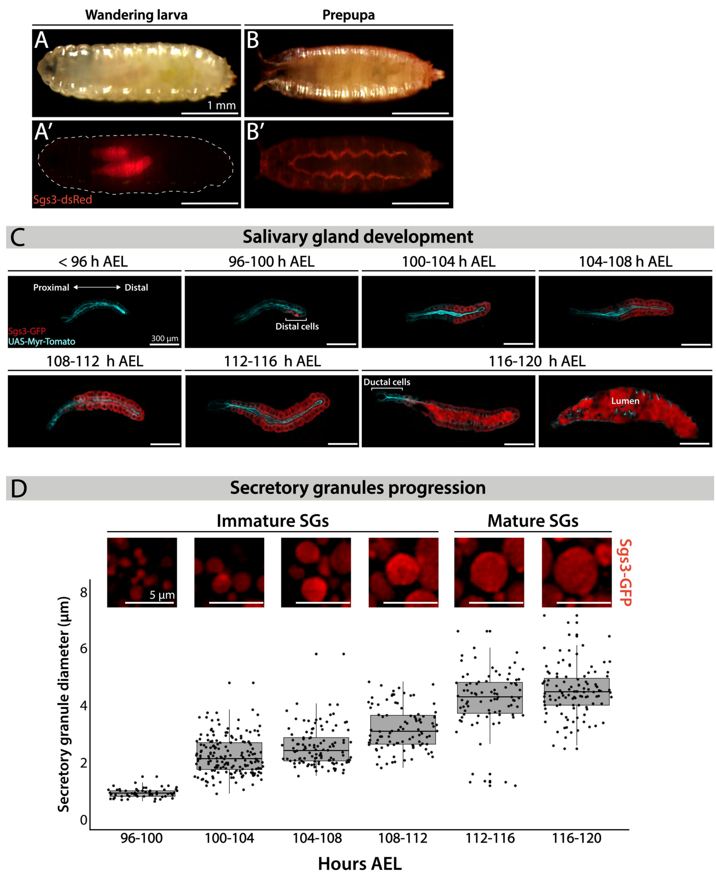


Figure 1.

Larval salivary gland as a model for regulated exocytosis.

Images of a wandering larva (A-A′) and a prepupa (B-B′) expressing Sgs3-dsRed and visualized under white light (A and B) or by epiflourescence (A′-B′). Sgs3-dsRed localized in salivary glands of wandering larvae (A′) and outside the puparium in prepupae (B′). Scale bar 1 mm. (C) Confocal images of unfixed salivary glands dissected from larvae at the indicated time intervals (h AEL= hours after egg laying). Sgs3-GFP is shown in red and the plasma membrane labelled with myr-Tomato (*sgs3-GFP, fkh-Gal4/ UAS-myr-Tomato*) is shown in cyan. At 96 h AEL, Sgs3 synthesis could be detected in the distal cells of the gland. Thereafter, Sgs3 expression gradually expanded proximally, and at 116 h AEL all salivary gland cells expressed Sgs3, with the exception of ductal cells. Exocytosis of SGs began by this time point, and Sgs3 could be detected in the gland lumen. At 120 h AEL concerted exocytosis of SGs has ended. Scale bar 300 μm. (D) Confocal images of SGs labelled with Sgs3-GFP; SG diameter distribution at each time interval is displayed below. Based on its diameter, SGs are classified as immature (diameter < 3 μm) or mature (diameter ≥ 3μm). Only SGs from distal cells were used for diameter determination. Data points for this graph are shown in **Supplementary Table 1**. For all time intervals analyzed n = 3, except for the 108-112 h AEL interval for which n = 4. “n” represents the number of salivary glands analyzed. Scale bar 5 μm.

Sgs3-GFP in salivary glands of prepupae (**Figure 2** and **Supplementary Figure 1B**). We used the salivary gland specific driver *forkhead*-Gal4 (*fkh*-Gal4) to induce the expression of RNAis against each of the eight subunits of the exocyst complex in larvae that also expressed Sgs3-GFP. We analyzed Sgs3-GFP distribution in wandering larvae and prepupae (**Figure 2A-B**), and observed that whereas control prepupae were able to expel Sgs3 outside the puparium (**Figure 2A**' and C), this process was blocked in most individuals expressing RNAis against any of the exocyst subunits, with Sgs3-GFP being detected inside salivary glands of prepupae (**Figure 2B**'-C and **Supplementary Figure 1B**). These observations suggest that all exocyst subunits are required for exocytosis of SGs.

Having established that knock-down of each of the exocyst subunits can block secretion of Sgs3, leading to retention of the mucin in the salivary glands (**Figure 2**), we investigated if exocytosis is actually impaired in these larvae. We dissected salivary glands of control and exocyst knock-down individuals at 116 h AEL, and analyzed them under the confocal microscope. Whereas control salivary gland cells were filled with mature SGs (**Figure 3A**), a more heterogeneous situation was found in salivary glands cells expressing RNAi against exocyst subunits, with some cells displaying mature SGs, and other, immature SGs, while even some displayed Sgs3-GFP in a mesh or network-like structure (**Figure 3B**). Therefore, we investigated if this mosaic phenotypic manifestation was due to variations in cell-to-cell Gal4-UAS activation and therefore RNAi expression, being an indication of potentially different functions of the exocyst complex in the secretory pathway of salivary gland cells. We performed RNAi-mediated knock-down of each of the eight exocyst subunits at different temperatures (29, 25, 21 and 19°C), that were accurately controlled in water baths, to obtain different levels of silencing and likely different phenotypic manifestations of exocyst complex downregulation. Whereas expression of a control RNAi resulted in mature SGs irrespectively of the temperature, expression of exocyst subunits RNAis at high temperatures (29°C) resulted, as a general rule, in phenotypes consistent with early arrest of the secretory pathway, since in most cells Sgs3-GFP was retained in a reticular structure or packed in immature SGs (**Figure 3C**, **Supplementary Figure 1A** and **Supplementary Table 3**). By lowering the temperature of RNAi expression, and therefore moderating silencing, the most severe phenotypes (early arrest of the secretory pathway) gradually became less prominent and simultaneously, the proportion of cells displaying immature SGs, and even mature SGs, became more noticeable (**Figure 3C** and **Supplementary Figure 1A** and **Supplementary Table 3**), suggesting that lower expression of RNAis allowed the progression of the secretory pathway. Notably, for each of the RNAis tested there was a temperature at which each of the three phenotypic outcomes could be clearly identified, although this particular temperature could be different for each RNAi, likely due to expression levels of the transgenes (**Figure 3C**, **Supplementary Figure 1A** and **Supplementary Table 3**). RT-qPCR confirmed that different culturing temperatures resulted in different degrees of silencing of exocyst subunit mRNA as shown in **Supplementary Figure 2** for *exo70* and *sec3* (**Supplementary Figure 2A**-**B**). Also, we found that different phenotypic outcomes resulting from expression of different RNAi transgenic lines that target the same subunit (*exo70)* are due to differences in the levels of mRNA downregulation generated by each particular ($exo70^{RNAiV}$ and $exo70^{RNAiBL}$), and that there is a correlation between the level of mRNA downregulation and strength of the phenotype observed (**Supplementary Figure 2C**).

To define more precisely the role of the exocyst, and rule out potential pleiotropic effects due to developmental defects derived from chronic exocyst downregulation, we made use of the Gal80 thermosensitive tool ($Gal80^{ts}$) ((Lee & Luo, 1999)). Larva were grown at a restrictive temperature (18°C) until they reached the 3er instar (120 hours). In this manner, the exocyst complex could be functional up until that developmental stage. Then, larvae were transfered to the permissive temperature (29°C) and salivary glands were dissected 36 hours later. We found that temporally-restricted expression of $sec3^{RNAi}$ or $sec15^{RNAi}$ phenocopied unrestricted expression of the same RNAis at 29°C (**Supplementary Figure 3**), indicating that the phenotypes obtained are not due to pleiotropic effects caused by developmentally unrestricted

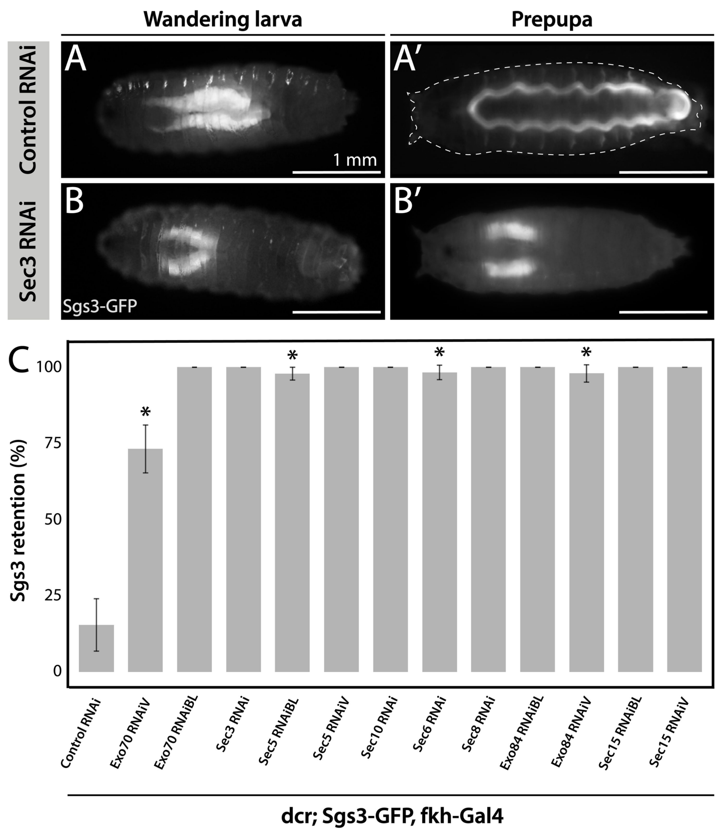


Figure 2.

The exocyst is required for Sgs3 secretion.

(A-B) Images of a larva and a prepupa expressing Sgs3-GFP, and visualized with epifluorescence. Sgs3-GFP was inside the salivary glands in control wandering larvae (A), and outside the puparium (A') in control prepupae (*sgs3-GFP*, *fkh-Gal4/UAS-cherry^{RNAi}*). Expression of *sec3* ^{RNAi} in salivary glands (*UAS-sec3^{RNAi}*; *sgs3-GFP, fkh-Gal4)* did not affect expression of Sgs3-GFP in larvae (B), but blocked Sgs3-GFP release outside the puparium, so the protein was retained inside the salivary glands (B'). This phenotypic manifestation is referred as "retention phenotype". Scale bar 1 mm. (C) Quantification of the penetrance of the retention phenotype in prepupae expressing the indicated RNAis. RNAis were expressed using *fkh-Gal4* and larvae were cultured at 29°C. All RNAis tested against exocyst complex subunits displayed a retention phenotype, with a penetrance significantly different from the control RNAi (*UAS-cherry^{RNAi}*) according to Likelihood ratio test followed by Tuckeýs test (p-value< 0.05). *cherry^{RNAi}* n= 7; *exo70^{RNAiV}* n= 11; *exo70^{RNAiBL}* n= 8; *sec3^{RNAi}* n= 6; *sec5^{RNAiBL}* n= 17; *sec5^{RNAiV}* n= 6; *sec10^{RNAi}* n= 9; *sec6^{RNAi}* n= 11; *sec8^{RNAi}*n= 5; *exo84^{RNAiBL}* n=5 ; *exo84^{RNAiV}*, n=6; *sec15^{RNAiBL}* n=3 ; *sec15^{RNAiV}* n=4. "n" represents the number of vials containing 20 to 30 prepupae per vial. For exocyst subunits with more than one RNAi line available, "BL" indicates a Bloomington Stock Center allele and "V" a Vienna Stock Center allele (see **Supplementary Table 2** for stock numbers).

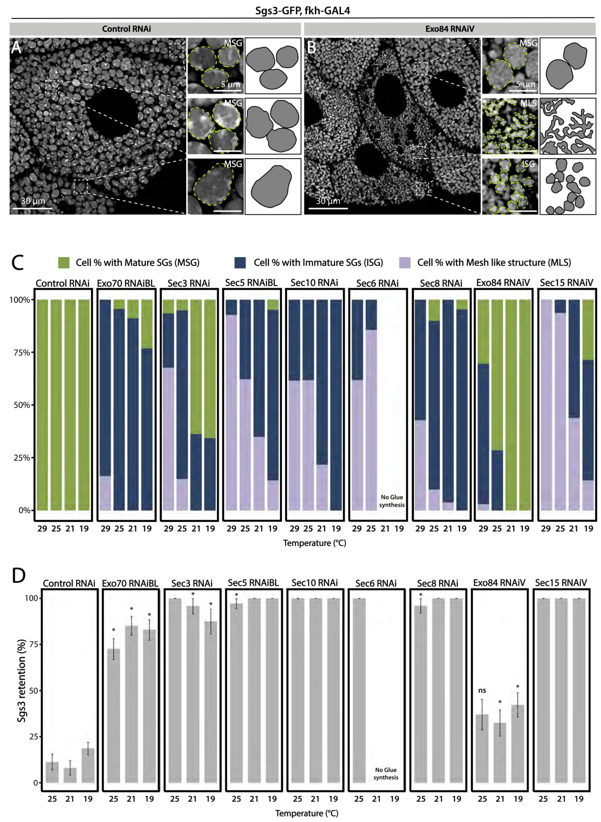


Figure 3.

Phenotypic manifestations of exocyst subunits silencing.

At the end of larval development (116-120 h AEL) salivary gland cells of control individuals (*sgs3-GFP*, *fkh-Gal4/ UAS-cherry*^{RNA}i) (A) were filled with mature SGs (insets). In cells expressing *exo84*^{RNAiV} (*UAS-exo84*^{RNAiV}*; sgs3-GFP*, *fkh-Gal4*) (B), three different phenotypes could be visualized in a single salivary gland: cells with mature SGs (MSG), cells with immature SGs (ISG) and cells with no SG, in which Sgs3 was retained in a mesh (MLS). Scale bar 30 µm in main panels, and 5 µm in insets. For didactic purposes, MSG, ISG and MLS were drawn on the right, next to the corresponding inset (C) Quantification of the penetrance of each of the three phenotypes observed upon downregulation of each of the exocyst subunits. Larvae were grown at four different temperatures (29, 25, 21 or 19°C) to achieve different levels of RNAi expression. “n” = Number of salivary glands analyzed; *control*^{RNAi} (*cherry*^{RNAi}) n = 4 (29 °C), n = 5 (25, 21 and 19 °C); *exo70*^{RNAiBL} n = 11 (29 °C), n = 7 (25, 21 °C), n = 6 (19 °C); *sec3*^{RNAi} n = 7 (29, 25 °C), n = 4 (21 °C), n = 9 (19 °C); *sec5*^{RNAiBL} n = 4 (29 °C), n = 12 (25 °C), n = 9 (21 °C), n = 6 (19 °C) ; *sec10*^{RNAi} n = 8 (29 °C), n = 6 (25, 19 °C), n = 7 (21 °C); *sec6*^{RNAi} n = 9 (29 °C), n = 4 (25 °C); *sec8*^{RNAi} n = 8 (29 °C), n = 6 (25, 19 °C), n = 7 (21°C); *exo84*^{RNAiV}n = 8 (29 °C), n = 4 (25, 19 °C), n = 5 (21 °C) ; *sec15*^{RNAiV} n = 4 (29, 25 °C), n = 6 (21 °C), n = 5 (19 °C). Raw data use to generate this graph is shown in **Supplementary Table 3**. (D) Quantification of the penetrance of the Sgs3-GFP retention phenotype in salivary glands of prepupae of the indicated genotypes. Only a few control individuals (*sgs3-GFP*, *fkh-Gal4/ UAS-cherry*^{RNA}i) displayed the retention phenotype. Downregulation of exocyst subunits provoked significant retention of Sgs3 inside the salivary glands irrespective to the temperature (25, 21 or 19 °C). Expression of *sec6*^{RNAi} at 21 or 19 °C resulted in no synthesis of Sgs3-GFP or Sgs3-dsRed, so the distribution of phenotypes was not assessed for this genotype at these temperatures. RNAis were expressed with *fkh*-Gal4. “n” = Number of vials with 20-30 larvae. *control*^{RNAi} (*cherry*^{RNAi}) n = 7 (25 °C), n = 9 (21 °C), n = 19 (19 °C); *exo70*^{RNAiBL} n = 13 (25 °C), n = 9 (21°C), n = 22 (19 °C); *sec3*^{RNAi} n = 9 (25 °C), n =6 (21, 19 °C); *sec5*^{RNAiBL} n = 9 (25 °C), n = 7 (21 °C), n = 12 (19 °C); *sec10*^{RNAi} n = 10 (25 °C), n = 5 (21 °C), n = 4 (19 °C); *sec6*^{RNAi} n = 10 (25 °C); *sec8*^{RNAi} n = 5 (25 °C), n = 8 (21 °C), n = 22 (19 °C); *exo84*^{RNAiV} n = 8 (25, 19 °C), n = 6 (21 °C) ; *sec15*^{RNAiV} n = 5 (25 °C), n = 9 (21 °C), n = 8 (19 °C). Statistical analysis was performed using a Likelihood ratio test followed by Tuckeýs test (“*”=p-value< 0.05). For those genotypes with 100% penetrance no statistical analysis was performed due to the lack of standard error. “ns”= not significant. Comparisons were made between RNAis for each of the temperatures analyzed.

downregulation of the exocyst. Finally, the MLS phenotype generated by expression of *sec15*^{RNAi} could be rescued by simultaneous expression of GFP-Sec15, supporting the notion that defective biogenesis of SGs was specifically provoked by *sec15* loss of function (**Supplementary Figure 4**).

A comprehensive analysis of cell polarity, as well as a number of general markers of cellular homeostasis (Supplementary Figures 5 and 6), ruled out that defects in SG biogenesis or maturation observed after knock-down of exocyst subunits could stem from potential secondary effects derived from poor cellular health, but rather reflect genuine functions of the exocyst complex in the secretory pathway. Along these lines, a recent report showed that apical polarity defects generated by loss of polarity protein Crumbs, do not affect or interfere with SG exocytosis (Lattner, Leng, Knust, Brankatschk, & Flores-Benitez, 2019), further supporting the notion that there are parallel pathways controlling cell polarity and SG biogenesis, maturation and exocytosis.

The fact that the three phenotypic outcomes (1-Sgs3-GFP retained in a mesh; 2-immature SG and 3-mature-not exocytosed SG) could be retrieved with appropriate silencing any of the eight subunits lead us to propose that the holocomplex, and not subcomplexes or individual subunits, could function several times along the secretory pathway, and that each of these activities could require different amounts of exocyst complex. Importantly, irrespectively of the temperature of expression of RNAis, retention of SGs in salivary glands cells was always significantly higher in exocyst knock-down individuals than in controls (**Figure 3D** and **Supplementary Figure 1B**), indicating that ultimately, the exocyst is required for SG exocytosis.

The exocyst complex is required for secretory granule biogenesis

We decided to characterize each of the three phenotypic manifestations of exocyst loss-of-function in more detail. The early-most manifestation of the requirement of the exocyst in the secretory pathway was the reticular or mesh-like phenotype obtained by strong silencing (29°C) any of the subunits (**Figure 4A**). This phenotype was reminiscent of mutants in which Sgs3 is retained at endoplasmic reticulum exit sites (ERES) or at ER-Golgi Complex (GC) contact sites (Burgess et al., 2011; Reynolds et al., 2019), suggesting that knock-down of the exocyst may provoke Sgs3 retention at the ER or GC, blocking SG biogenesis. Indeed, we found that RNAi-mediated silencing of *sec15*,*sec3* or *sec10* provoked Sgs3 retention in the ER since it colocalized with ER markers, as indicated by Pearsońs coefficient, whereas in control larvae Sgs3-GFP was inside SGs, at a comparable developmental stage (**Figure 4B-G** and Supplementary Figures 7 and 8).

We analyzed in detail GFP-Sec15 subcellular localization, and found that associated closely with the *trans*-Golgi marker RFP-Golgi, but not to the ER marker KDEL-RFP, in salivary glands just prior to SG biogenesis (**Figure 5** and Supplementary Movie 1). This association was lost after the onset SG biogenesis, suggesting that the exocyst associates with the GC at the specific developmental stage when Sgs3 transits through that organelle (**Figure 5B-E**). Three-dimensional reconstruction of Sec15-Golgi *foci* confirmed the spatial association with the exocyst (**Figure 5E** and Supplementary Movie 2-3). In line with this, we found that exocyst silencing under conditions that block Sgs3-GFP in the ER (29°C), both, *cis* and *trans*-CG structures were severely affected (**Figure 6** and **Supplementary Figure 9**). This suggests that at this stage of salivary gland development the exocyst complex localizes to the GC, where it is required to maintain *cis*- and *trans*-GC morphology and allow the correct transport of Sgs3 from the ER through the GC (**Figure 6E-F**).

Role of the exocyst in secretory granule maturation: Homotypic fusion

Given that appropriate silencing conditions of any of the exocyst subunits can result in accumulation of immature SGs (**Figure 7A**), we set out to investigate a potential role of the exocyst in SG maturation. SG maturation is a multifaceted process that involves, among other

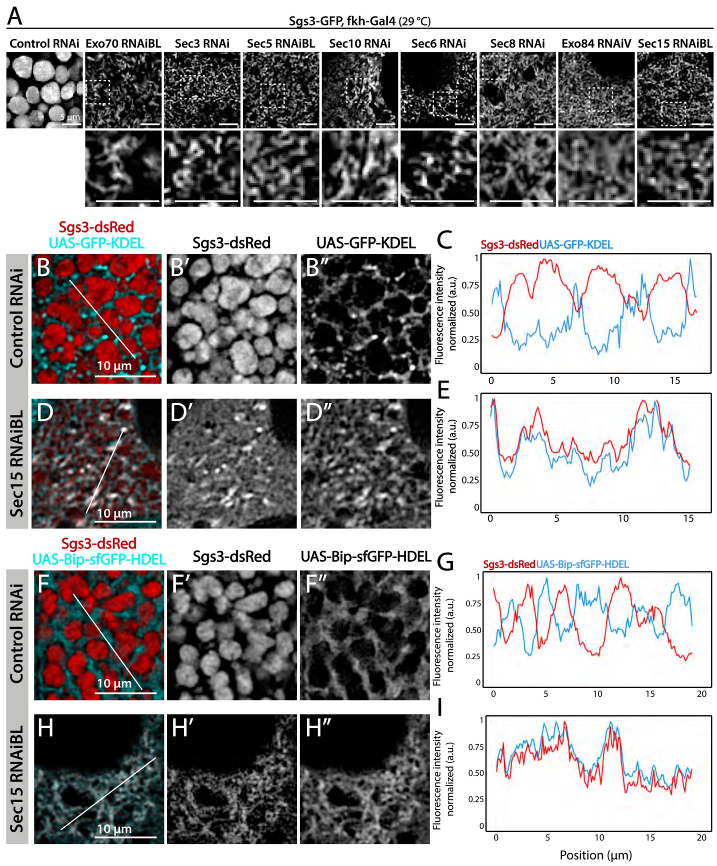


Figure 4.

The exocyst is required for Sgs3-GFP exit from ER and SG biogenesis.

Confocal images of unfixed salivary glands of the indicated genotypes. Larvae were grown at 29 °C to achieve maximal activation of the Gal4-UAS system, and maximal downregulation of exocyst subunits. (A) In control salivary glands (*sgs3-GFP*, *fkh-Gal4/ UAS-cherry^{RNAi}*) 116-120 AEL, Sgs3-GFP was packed in mature SGs. Salivary glands expressing RNAis against each of the exocyst subunits, where Sgs3-GFP exhibited a reticular distribution are shown. SGs were not formed in these cells. Scale bar 5 µm. The ER, labeled with GFP-KDEL (*UAS-GFP-KDEL*), (B and D) or Bip-sfGFP-HDEL (*UAS-Bip-sfGFP-HDEL*), (F and H) distributed in between SGs in control salivary glands (B and F); in sec15^{RNAi} salivary glands (*fkh-Gal4/ UAS-sec15^{RNAiBL}*) SGs did not form and the Sgs3-dsRed signal overlapped with the ER markers (D and H). (C, E, G and I) Two-dimensional line scans of fluorescence intensity across the white lines in panels B, D, F and H of Sgs3-dsRed and the ER markers. In all cases transgenes were expressed using *fkh*-Gal4. Scale bar 10 µm.

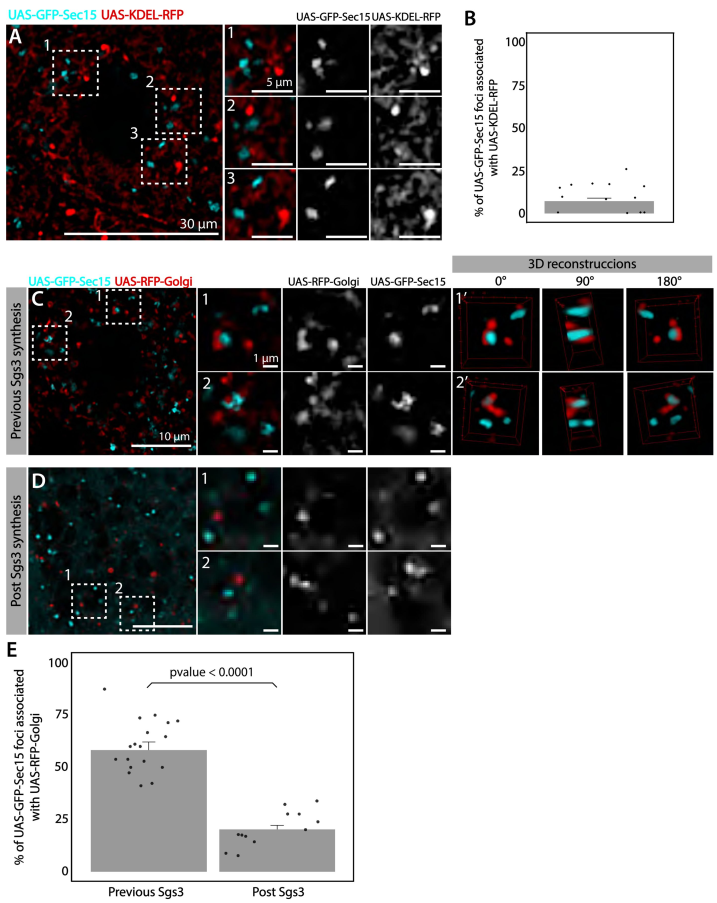


Figure 5.

The exocyst localizes at the Golgi complex before Sgs3 synthesis.

Confocal images of unfixed salivary glands expressing GFP-Sec15 and (A) the ER marker KDEL-RFP or (C) the trans-Golgi marker RFP-Golgi. Sec15 foci did not localize at or associate with the ER (A-B). (C) Sec15 foci were found in close association with trans-Golgi complex cisternae. Examples of association events are shown in insets 1 and 2, including different angles of three-dimensional reconstruction stacks. (D) Sec15 foci and trans-Golgi complex association was lost after Sgs3 synthesis has begun. (E) Quantification of Sec15 foci-Golgi complex association events before and after the onset of Sgs3 synthesis. (Wald Test, p-value < 0.05). “n” represents the number of salivary glands analyzed, n=4.

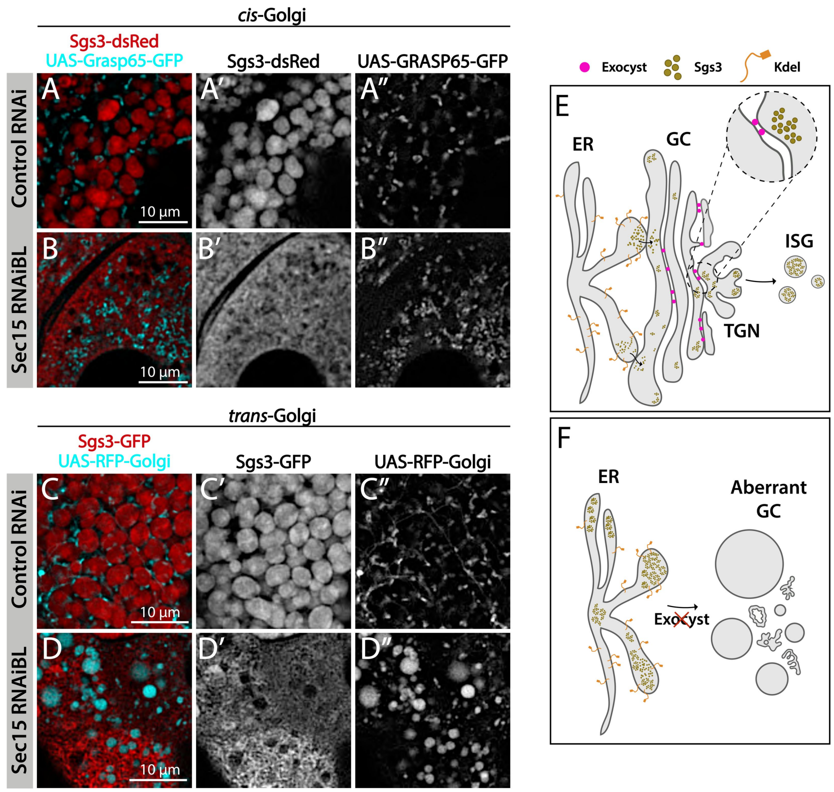


Figure 6.

The exocyst is required to maintain normal Golgi complex morphology.

Confocal images of unfixed salivary glands of the indicated genotypes. Larvae were grown at 29 °C to achieve maximal activation of the Gal4-UAS system and maximal downregulation of exocyst subunits. In control larvae (*fkh-Gal4/ UAS-white^{RNAi}*), Sgs3-dsRed (A) or Sgs3-GFP (C) were in mature SGs. In sec15^{RNAi} salivary glands (*fkh-Gal4/ UAS-sec15^{RNAiBL}*) Sgs3 was retained in a mesh-like structure (B and D), also shown in **Figure 4**. (A, B) The *cis*-Golgi complex was labelled with Grasp65-GFP (*UAS-Grasp65-GFP*), (and the *trans*-Golgi with RFP-Golgi (*UAS-RFP-Golgi*), (C, D). The morphology of the *cis*- and *trans*-Golgi complexes changed dramatically in sec15-knockdown cells (B″, D″), in comparison to controls (A″, C″). Transgenes were expressed with *fkh-Gal4*. Scale bar 10 µm. (E) Model of Sgs3 transit from the ER through the *cis*- and trans-Golgi complex to sprouting SGs from the *trans*-Golgi complex. (F) The exocyst is needed for tethering the Golgi complex cisternae and to support Golgi complex structure. In the absence of the exocyst Golgi cisternae disconnect, *cis*- and trans-Golgi become dysfunctional, resulting in Sgs3 retention at the ER.

events, homotypic fusion between immature SGs (Du et al., 2016; Neuman & Bashirullah, 2018). We found that GFP-Sec15 often localized in discrete *foci* in between immature SGs (75% of total GFP-Sec15), but this localization drops dramatically when SGs have undergone maturation (15%) (**Figure 7B**, C and Supplementary Movie 4-5). This transient localization of the GFP-Sec15 supports the notion of a role of the exocyst in SG homofusion, which was confirmed by live imaging of salivary glands *ex-vivo* (**Figure 7D** and Supplementary Movie 6). In support of this notion, when GFP-Sec15 was overexpressed at 25 °C, unusually large SGs of up to 20 µm in diameter could be detected, a size never observed in control salivary glands overexpressing GFP alone (**Figure 7E-H**), indicating that overexpression of the Sec15 subunit alone is sufficient to induce homotypic fusion between SGs, which is in agreement with reports that indicate that Sec15 functions as a seed for exocyst complex assembly (Escrevente, Bento-Lopes, Ramalho, & Barral, 2021; Guo, Roth, Walch-Solimena, & Novick, 1999). Sec15 overexpression is expected to induce the formation of of exocyst holocomplexes, provoking excessive homofusion among SGs. In contrast, overexpression of Sec8, which is not expected to induce the formation of the whole complex, did not have an effect on homotypic fusion of SGs (**Figure 7F, H**).

The characteristic localization of GFP-Sec15 *foci* in between adjacent immature SGs, the fact that Sec15 overexpression results in oversized SGs, and the observation that downregulation of any subunit of the exocyst complex at appropriate levels results in accumulation of immature SGs, weigh in favor of the notion that the exocyst plays a critical role in SG homotypic fusion.

Role of the exocyst in secretory granule maturation: acquisition of membrane proteins

Besides homotypic fusion, maturation of SGs involves the incorporation of specific proteins that are required for homotypic fusion, apical navigation or fusion with the apical plasma membrane. The mechanisms by which these maturation factors associate with SG are not well understood. One of such proteins is the calcium sensor Synaptotagmin-1 (Syt-1), which localized at the basolateral membrane of salivary gland cells before SG biogenesis (96 h AEL) (**Supplementary Figure 10A**) and later, became detectable on the membrane of nascent SGs (diameter<1µm), prior to homotypic fusion (**Figure 8A** and **Supplementary Figure 10B**). As SGs mature, the presence of Syt-1 on SGs became more prominent, with a sharp increase after SG homofusion (**Supplementary Figure 10C**-**D**), suggesting that recruitment of Syt-1 to SGs continued after homotypic fusion had occurred (**Figure 8M**). Downregulation of the exocyst subunits Sec5 or Sec3 to levels that generate immature SGs, significantly reduced the presence of Syt-1 on their membranes, as compared to immature SGs of control salivary glands (**Figure 8A-C** and **Supplementary Figure 11A**-**C**). Interestingly, weaker knock-down, which allowed the formation of mature SGs, improved but not completely restored Syt-1 recruitment, compared to control salivary cells (**Supplementary Figure 11D**-**F**), indicating that Syt-1 is recruited to mature SGs in an exocyst-dependent manner (**Figure 8M**).

CD63, the *Drosophila* homolog of mammalian Tsp29Fa, is another protein required for SG maturation. CD63 localizes at the apical plasma membrane before SG formation and reaches the membrane of SGs through endosomal retrograde trafficking (**Supplementary Figure 10E**), (Ma et al., 2020). We investigated if the exocyst participates in recruitment of CD63 to SGs, and found that CD63 could be readily detected at the membrane of 1µm; 3µm or 5µm SGs (**Figure 8D** and **Supplementary Figure 10F**-**H**), while upon Sec5 or Sec3 downregulation CD63 was significantly reduced on immature SGs (**Figure 8D-F** and **Supplementary Figure 11G**-**I**). Interestingly, milder reduction of Sec3 expression, under conditions that allow the formation of mature SGs, did not affect CD63 recruitment to SGs (**Supplementary Figure 11J**-**L**), indicating that CD63 is recruited to SGs before homotypic fusion in an exocyst-dependent manner and that, unlike to Syt-1, recruitment ceases after homotypic fusion has occurred (**Figure 8M**).

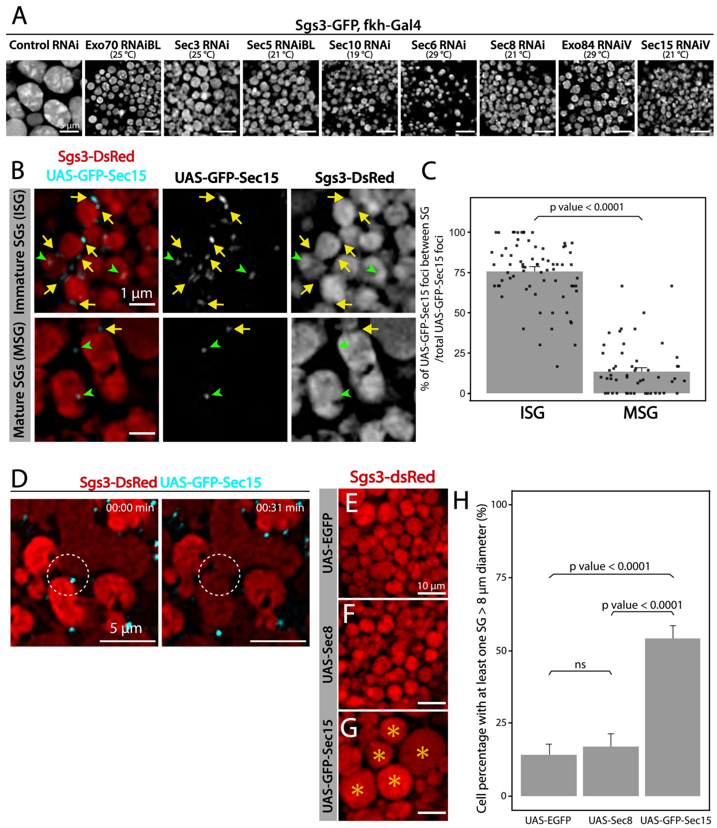


Figure 7.

The exocyst is required for secretory granule homotypic fusion.

(A) Confocal images of unfixed salivary glands of the indicated genotypes. In control salivary glands (*sgs3-GFP*, *fkh-Gal4/ UAS-cherry^{RNA}*i), Sgs3-GFP was packed in mature SGs. RNAis targeting any of the subunits of the exocyst were expressed at the indicated temperatures, giving rise to salivary cells with immature granules. Scale bar 5 µm. (B) GFP-Sec15 (cyan) and Sgs3-dsRed (red) were expressed in salivary glands (*sgs3-dsRed/ UAS-GFP-sec15; fkh-Gal4*) at 18 °C. GFP-Sec15 foci were mostly found in between SGs (yellow arrows) in cells bearing immature SGs (upper panel) and not in cells with mature SGs (lower panel), in which most foci did not localize in between SGs (green arrowheads). Scale bar 1 µm. (C) Quantification of Sec15 foci in between SGs relative to the total number of Sec15 foci (%) (Wald Test, p-value < 0.05). Eight salivary glands with ISGs and 5 salivary glands with MSGs were analyzed. (D) Still panels of two different frames of Supplementary movie 6 showing that during a homotypic fusion event, GFP-Sec15 accumulated precisely at the contact site between two neighboring SGs (dotted circle); scale bar 5 µm. (E-G) Expression of EGFP (*sgs3-dsRed; UAS-EGFP, fkh-Gal4*) (E) or Sec8 (*sgs3-dsRed/ UAS-Sec8; fkh-Gal4*) (F) did not affect SG size (red). Expression of GFP-Sec15 (G) (*sgs3-dsRed/ UAS -GFP-sec15; fkh-Gal4*) generated giant SGs (asterisks). Scale bar 10 µm. (H) Quantification of the percentage of salivary gland cells with at least one SG larger than 8 µm diameter; (Likelihood ratio test followed by Tuckeýs test, p-value < 0.05). “n” represents the number of salivary glands analyzed. UAS-EGFP n=7; UAS-Sec8 n=5; UAS-GFP-Sec15 n=5. Transgenes were expressed using *fkh-Gal4.* “ns”= not significant.

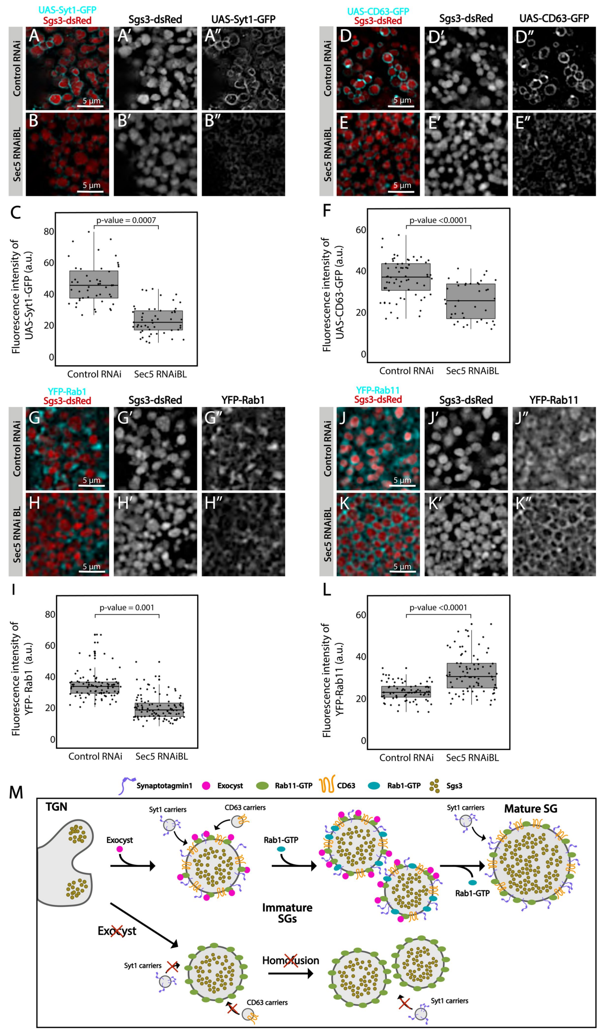


Figure 8.

The exocyst mediates the acquisition of secretory granule maturation factors.

Confocal images of unfixed salivary glands of the indicated genotypes. Larvae were grown at 21 °C to reduce the activity of the Gal4-UAS system, and to generate a maximal proportion of cells with immature SGs. (A-B) Recruitment of Syt1-GFP (*UAS-Syt1-GFP*); of (D-E) CD63-GFP (*UAS-CD63-GFP*); of (G-H) YFP-Rab1 and of (J-K) YFP-Rab11 was analyzed in control salivary glands expressing *white*^{RNAi} (*fkh-Gal4/ UAS-white*^{RNAi}) and in salivary glands expressing *sec5*^{RNAiBL} (*fkh-Gal4/ UAS-sec5*^{RNAiBL}). Fluorescent intensity around SGs of each of the analyzed maturation factors was quantified using the ImageJ software and plotted (C, F, I and L). Comparison of fluorescent intensity among genotypes and statistical analysis was performed using one-way analysis of variance (ANOVA). “n” represents the number of salivary glands: (C) *control RNAi* n= 4, *sec5*^{RNAiBL} n= 5; (F) *control RNAi* n= 5, *sec5*^{RNAiBL}n=4; (I) *control RNAi* n= 6, *sec5*^{RNAiBL} n= 9; (L) *control RNAi* n= 11, *sec5*^{RNAiBL} n= 8. Transgenes were expressed using *fkh-Gal4*. Scale bar 5 µm. (M) Proposed model of exocyst-dependent SG homotypic fusion and maturation. The exocyst complex is required for homotypic fusion between immature SGs. Immature SGs incorporate Syt-1, CD63 and Rab1 in an exocyst-dependent manner. Syt-1 continues to be recruited to SGs after homotypic fusion. The exocyst complex also inhibits the incorporation of an excess of Rab11 around SGs.

The small GTPases Rab11 and Rab1 were reported to be required for SG maturation. Whereas Rab11 associates to SGs independently of their size, Rab1 was found transiently on immature SGs only (Ma & Brill, 2021a; Neuman et al., 2021), and moreover, Rab1 recruitment to SGs depends on Rab11 (Neuman et al., 2021). Given that Rab11 is a stable component of SGs and Rab1 is not, we reasoned that other players, perhaps the exocyst, might be involved in Rab1 association or dissociation from the SG membrane. In fact, silencing of Sec5 under conditions that allowed the formation of immature SGs, provoked significant reduction of Rab1-YFP on the SG membrane (**Figure 8G-I**), suggesting that the exocyst participated in Rab1 recruitment (**Figure 8M**).

Remarkably, immature SGs that resulted from Sec5 or Sec3 knock-down displayed higher-than-normal levels of YFP-Rab11 around them (**Figure 8J-L** and **Supplementary Figure 11M**-**O**), suggesting that the exocyst is a negative regulator of Rab11 recruitment. This observation is in apparent contradiction with previous reports by others and us that indicate that Rab11 is required for SG maturation (de la Riva-Carrasco et al., 2021; Ma & Brill, 2021a; Neuman et al., 2021). Noteworthy, overexpression of a constitutively active form of Rab11 (UAS-YFP-Rab11^{CA}), which was readily recruited to SGs, also provoked an arrest of SG maturation, thereby phenocopying exocyst complex knock-down (**Supplementary Figure 12A**-**C**). Thus, during maturation of SGs, the levels of active Rab11 need to be precisely regulated and this is achieved, at least in part, by the exocyst (**Figure 8M** and **Supplementary Figure 12F**).

Given that the exocyst is an effector of different Rab-GTPases during vesicle exocytosis (Guo et al., 1999; Novick et al., 1995; S. Wu, Mehta, Pichaud, Bellen, & Quiocho, 2005), we next investigated if Rab11 is required for recruiting the exocyst to SGs. Indeed, we found that Sec15 failed to localize on SGs following knock-down of Rab11 (**Supplementary Figure 12D**-**E**), indicating a crucial role of Rab11 in recruitment of Sec15, and probably of the whole exocyst to SGs.

Overall, the results described in this section suggest that Rab11 recruits Sec15, and perhaps the whole exocyst, to immature SGs to allow homotypic fusion and maturation (**Supplementary Figure 12D**-**E**) while in turn, the exocyts limits the levels and/or activity of Rab11 on SGs (**Figure 8J-L** and **Supplementary Figure 11M**-**R**), as too much Rab11 is apparently detrimental for SG maturation (**Supplementary Figure 12A**-**C**). We propose that a single-negative feedback loop precisely regulates Rab11 and exocyst complex activity/levels, thus controlling recruitment of maturation factors such as Syt-1, CD63 and Rab1, and therefore, the outcome of SG maturation (**Supplementary Figure 12F**).

The exocyst is required for secretory granule fusion with the apical plasma membrane

Under low silencing conditions of exocyst subunits the prevalent phenotype was mature SGs retained in salivary gland cells (**Figure 3D**, **Figure 9A**, and **Supplementary Figure 1B**), suggesting a function of the complex in SG fusion with the apical plasma membrane. The process of SG-plasma membrane fusion can be assessed by visualizing the incorporation of plasma membrane-specific components to the membrane of SGs (de la Riva-Carrasco et al., 2021; Rousso et al., 2016; Tran et al., 2015). Phosphatidylinositol 4,5-bisphosphate (PI(4,5)P_2) is a lipid of the inner leaflet of the apical plasma membrane, which is absent in endomembranes (Phan et al., 2019). Using a fluorophore-based reporter of PI(4,5)P_2 the SGs that have fused with the plasma membrane can be distinguished from those that have not. Using this approach we found that silencing of Exo70 impaired fusion with the plasma membrane (**Figure 9B-D**). Consistent with a role of the exocyst in fusion of SGs with the plasma membrane, at this stage of development GFP-Sec15 was no longer detected GFP-Sec15 was detected on SGs at the sites of contact with the plasma membrane (**Figure 9E-G** and Supplementary Movie 7). These data indicate that during regulated exocytosis, the exocyst complex is required for fusion of SGs to the apical plasma membrane, possibly acting as a tethering complex (**Figure 9H-I**).

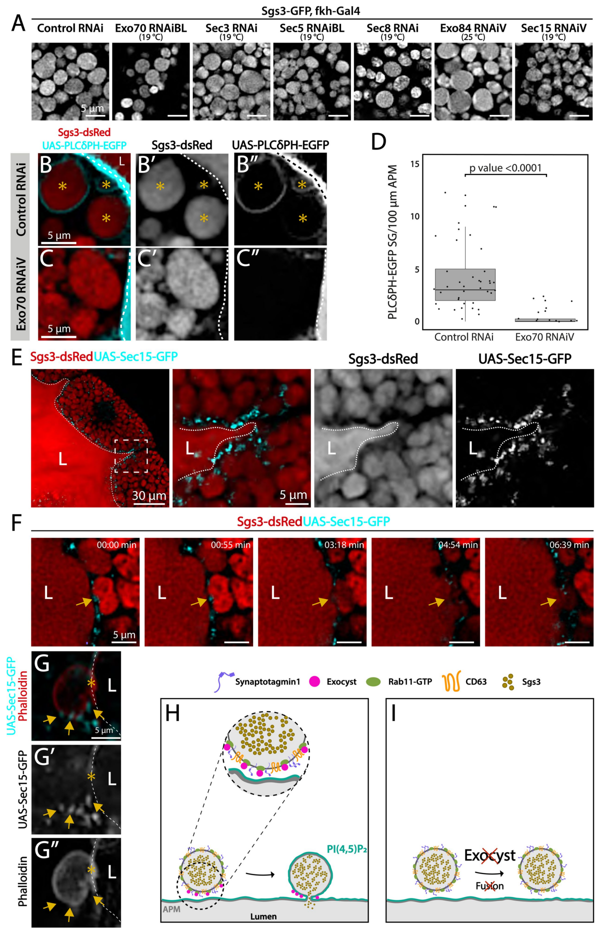


Figure 9.

The exocyst is required for secretory granule fusion with the plasma membrane during regulated exocytosis.

Confocal images of unfixed salivary glands of the indicated genotypes. Larvae were grown at the indicated temperatures to attain levels of RNAi-mediated silencing that bring about maximal proportion of cells with mature, exocytosis incompetent SGs. (A) Sgs3-GFP localized within SGs. Control SGs (*sgs3-GFP*, *fkh-Gal4/ UAS-cherry*^{RNAi}) were indistinguishable from SGs of salivary cells in which a subunit of the exocyst has been knocked-down. Scale bar 5 µm. (B) In control salivary glands (*fkh-Gal4/ UAS-white*^{RNAi}), the PI(4,5)P_2 reporter *UAS-PLCγ-PH-GFP* labels the plasma membrane (dotted line) and also the SGs that have already fused with the plasma membrane (asterisks). (C) SGs of cells expressing *exo70*^{RNAiV} (*UAS-exo70*^{RNAiV}*; fkh-Gal4*) were not labeled with the reporter, indicating that these SGs failed to fuse with the plasma membrane. Scale bar 5 µm. (D) The number of mature SGs positive for PLCγ-PH-EGFP per 100 µm of linear plasma membrane was quantified in the indicated genotypes; Exo70 knock-down reduced SG-plasma membrane fusion (Wald test (p value<0.05); 7 salivary glands per genotype were analyzed. (E) During SG exocytosis the exocyst complex, labelled with GFP-Sec15 (cyan) localized as dots in contact sites between SGs and the apical plasma membrane (dotted line) (*sgs3-dsRed/ UAS-GFP-sec15; fkh-Gal4*). (F) Still panels of Supplementary movie 2 showing a fusion event between a mature SG and the plasma membrane (arrow); a dot of GFP-Sec15 indicating the position of the exocyst (arrow) was positioned just at the site where fusion was taking place. Scale bar 5 µm. (G) Confocal image of a fixed salivary gland. A mature SG that has fused with the plasma membrane (dotted line), and thus became labelled with phalloidin, displayed dots of GFP-Sec15 on its side (arrow), just next to the fusion point (asterisk). Transgenes were expressed with *fkh-Gal4*. The salivary gland lumen is indicated with “L”. Scale bar 5 µm. (H) Model of the role of the exocyst complex during SG-plasma membrane fusion during regulated exocytosis. The exocyst sits on the membrane of the SG, and tethers the granule to the plasma membrane, favoring the action of fusion molecules. (I) Upon loss of the exocyst complex, mature SGs cannot contact the plasma membrane and fusion does not occur.

Overall, by utilizing the *Drosophila* larval salivary gland, we have made a comprehensive analysis of the role of the exocyst complex in the pathway of regulated exocytosis. We found that the exocyst is critically required for biogenesis of SGs, for their maturation and homotypic fusion and for mediating fusion between SGs and the plasma membrane.

Discussion

The exocyst was initially identified as crucial for secretion in yeast (Novick et al., 1980; TerBush et al., 1996) and later on, for basolateral trafficking in animal epithelial cells (Grindstaff et al., 1998; Lipschutz et al., 2000); (Langevin et al., 2005), with implications in various cellular processes, including cell migration, cytokinesis, ciliogenesis and autophagy (Bodemann et al., 2011; Park et al., 2010; Rogers et al., 2004; Thapa et al., 2012). Recently, an in-depth study of the mammalian exocyst was performed, showing that each of the eight subunits of the complex are essential for constitutive secretion of soluble proteins (Pereira et al., 2023). However, fewer is known about the function of the exocyst in regulated exocytosis. A recent work indicates that, in cultured pancreatic beta cells, the exocyst mediates tethering of insulin-containing granules to the plasma membrane and to the cortical F-actin network, being required for exocytosis of a subset of these granules (Zhao et al., 2023). Previous studies had implicated specific subunits of the complex in insulin-stimulated exocytosis of the glucose transporter Glut4 in cultured adipocytes (Inoue, Chang, Hwang, Chiang, & Saltiel, 2003), as well as in skeletal muscle cells (Fujimoto et al., 2019). In *Drosophila*, some subunits of the complex (Sec3, Sec5, Sec6, Sec8 and Sec15) were recently suggested to be required for SG development (Ma et al., 2020), although earlier, a role for the Sec10 in salivary gland exocytosis had been disregarded (Andrews, Zhang, Trotta, & Broadie, 2002). Therefore, a role of the exocyst in regulated exocytosis is less clear. In this study, we not only demonstrate that the eight subunits of the complex are required for regulated exocytosis of SGs of *Drosophila* larval salivary gland, but also that the holocomplex participates in multiple steps along the secretory pathway.

Most studies of the secretory pathway rely on a bimodal readout: Intracellular retention of SGs *versus* exocytosis of SG content. These approaches set a limit to our understanding of the specific functions that regulators and effectors exert on the exocytic pathway. By using fluorescently labeled versions of a SG cargo protein in *Drosophila* salivary glands, Sgs3, we have shown that knock-down of any of the exocyst subunits can bring about three distinct phenotypic outcomes: 1-Impairment of SG biogenesis, 2-impairment of SG homotypic fusion and maturation, or 3-impairment of SG fusion with the apical plasma membrane. Noteworthy, the frequency at which these defects occurs depends on the extent of exocyst downregulation, implying that each of the three functions depends on different levels of the complex: SG-plasma membrane fusion is highly sensitive to even slight reductions of exocyst subunits levels; SG maturation requires intermediate levels of the complex, and SG biogenesis seems to be the most robust of the three processes, and strong reduction of exocyst levels is required to bring about this defect.

Although exocyst subcomplexes or individual subunits have been suggested to play specific roles in cell biology (Mehta et al., 2005), it is generally believed that the functions of the exocyst are carried out by the holocomplex (Ahmed et al., 2018). Studies analyzing the requirement of all eight subunits in a single biological process are not abundant. In *Drosophila*, a study was performed in which the eight subunits were found to be required for developmentally regulated autophagy, during salivary cell death, but not for starvation-induced autophagy in the fat body (Tracy et al., 2016) and also for proper synaptic development at the neuromuscular junction (Kang et al., 2024). However, in other biological settings some subunits of the exocyst (Sec3, Sec5, Sec6, Sec8, and Sec10) but not others (Sec15, Exo70 and Exo84) were shown to be specifically required for general as well as specific autophagy in yeast (Singh et al., 2019). In mammalian cells, Exo84- and Sec5-induced exocyst complex activation have opposing roles on autophagy machinery activation (Bodemann et al., 2011). Our systematic analysis of the requirement of each of the exocyst subunits in biogenesis, maturation and exocytosis of SGs led us to conclude that all three biological processes are carried out by the holocomplex, and not by individual subunits or subcomplexes.

Consistent with the three biological functions that the exocyst exerts in the secretory pathway, we found that its subcellular localization during salivary gland development is dynamic. Immediately after the onset of Sgs3 synthesis, the exocyst localized at the Golgi complex; later, when SG biogenesis has begun the exocyst was mostly present at the fusion point between immature SGs, and finally it localized on mature SGs in close proximity to the apical plasma membrane. These different subcellular localizations are coherent with the three phenotypes that we have observed after precise modulation of exocyst levels. Exocyst localization at the trans-Golgi network and at the plasma membrane has been reported before in mammalian cells (Yeaman, Grindstaff, Wright, & Nelson, 2001), and although it was suggested in that study that the exocyst might be required at several steps of the secretory pathway from the TGN to the plasma membrane, the idea was not further explored.

By analyzing SG-plasma membrane fusion markers, we have shown that, paralleling the requirement of the exocyst in constitutive exocytosis, the complex participates in SG fusion with the plasma membrane in regulated exocytosis as well. Moreover, we have found that the Sec15 subunit localizes exactly at contact sites between SGs and the plasma membrane, further supporting the notion that the exocyst plays a role in this fusion process. It is accepted that the function exerted by the exocyst in the fusion between SGs and the plasma membrane depends on Ral GTPases (Brymora, Valova, Larsen, Roufogalis, & Robinson, 2001; Wang, Li, & Sugita, 2004). RalA is the only Ral GTPase described so far in *Drosophila*, and we have recently reported that RalA is involved in SG-plasma membrane fusion in the salivary gland (de la Riva-Carrasco et al., 2021). Therefore, it seems likely that, paralleling constitutive exocytosis, the interaction between RalA and the exocyst is critical for tethering SGs to the plasma membrane during regulated exocytosis as well. The molecular components regulating tethering and fusion of SGs to the plasma membrane remain largely unknown. Perhaps, this is because the proteins involved in these processes participate in earlier steps of the secretory pathway as well. Our temperature-dependent manipulation of the expression of exocyst subunits, in combination with differential requirements of the complex in terms of quantity/activity in each of the processes was instrumental for uncovering three different discrete steps in which the exocyst plays a role along the regulated secretory pathway.

Proteins that will be secreted are co-translationally translocated to the ER, and then transported into the cis-region of the GC in COPII-coated vesicles that are 60–90 nm in diameter and therefore, sufficient to accommodate most membrane and secreted molecules (Raote & Malhotra, 2021). Larger cargos, such as collagen and mucins might use alternative ER-cis-GC communication mechanism that are independent of vesicular carriers. Specifically, direct connections between the ER and cis-GC are formed in a Tango1 and COPII dependent manner (Reynolds et al., 2019)(Yang, Feng, & Pastor-Pareja, 2024). Disruption of these connections not only affects SG formation, but also has profound impact on GC structure (Bard & Malhotra, 2006; Rios-Barrera, Sigurbjornsdottir, Baer, & Leptin, 2017). We found that strong silencing of any of the exocyst subunits results in retention of Sgs3 in the ER, abrogation of SG biogenesis, and alteration of normal morphology of the GC. In *Drosophila*, as well as in mammalian cells, Golgi complex is polarized in *cis* and *trans*-cisternae that are held together by tethering complexed of the CATCHR family. Disruption of these connections result in altered Golgi complex morphology, reflected in fragmentation and swelling of *cisternae* and impairment of the secretory pathway (D'Souza, Taher, & Lupashin, 2020; Khakurel & Lupashin, 2023; Liu et al., 2019). The fact that exocyst silencing not only affected SG formation, but also had profound impact on GC structure, together with the observation that Sec15 specifically associated with the Golgi complex but not with the ER before the onset of SG biogenesis, argues in favor of a role of this complex as a tethering complex between cisternae. We propose that the exocyst might be redundant with other CATCHR complexes in this function since only severe downregulation of exocyst subunits expression can manifest this phenotype.

Maturation of SGs is a multidimensional process that involves homotypic fusion, acidification, cargo condensation, and acquisition of membrane proteins that will steer SGs to the apical plasma membrane and contribute to the recognition, tethering and fusion (Boda et al., 2023; Ji et al., 2018; Nagy et al., 2022; Syed, Zhang, Tran, Bleck, & Ten Hagen, 2022). Previously, the exocyst was shown to be required for maturation of Weibel-Palade bodies (Sharda et al., 2020). In the current work, we have shown that the exocyst participates in several aspects of SG maturation. One of such processes is homotypic fusion of immature SGs. We found that an adequate extent of downregulation of the expression of any of the exocyst subunits leads to accumulation of immature SGs. Furthermore, consistent with a function of the exocyst in SG homofusion, we found that Sec15 localizes preferentially at the fusion point between adjacent immature SGs. In support of this notion, we found that overexpression of Sec15 results in unusually large granules, likely derived from uncontrolled homotypic fusion. Therefore, our data suggest that the exocyst complex is the tethering factor responsible to bring immature SGs in close proximity to enable fusion.

Besides SG growth by homotypic fusion, several factors are recruited to the maturing SG, including the transmembrane proteins CD63 and Syt-1. Our temperature-dependent genetic manipulations revealed that the addition of Syt-1 to SGs occurs immediately after they have emerged from the TGN, and addition of this protein continues after they have attained a mature size. Syt-1 localizes at the basolateral membrane of salivary gland cells before SG biogenesis, and it is not known how Syt-1 reaches the maturing SGs. Given that the exocyst is required for loading SGs with Syt-1, one possibility is that a vesicular carrier transports Syt-1 from the basolateral membrane to nascent and maturing SGs, and that the contact between SGs and these vesicles is mediated by the exocyst. CD63, which is also required for SG maturation, localizes instead at the apical plasma membrane before SG biogenesis, and reaches the SGs from the endosomal retrograde pathway through endosomal tubes (Ma et al., 2020). We found that cells in which the exocyst has been knocked-down display reduced levels of CD63 around SGs, suggesting that the exocyst might contribute to the contact between endosomal tubes and SGs as well. Based on our analysis, this maturation event should take place mostly before, and not after, homotypic fusion. Thus, our results support the notion that the exocyst complex might contribute to tethering maturing SGs to different types of membrane-bound carriers that transport maturation factors like Syt-1 and CD63. These observations expand the notion that a crosstalk between the secretory and endosomal pathways occurs (Ma & Brill, 2021a; Ma et al., 2020; Papandreou & Tavernarakis, 2020), and that the exocyst is a critical factor linking the two pathways.

Vesicle maturation, either in the endocytic or the exocytic pathways, usually involves changes in vesicle-bound Rab proteins (Ailion et al., 2014; Ma & Brill, 2021a; Thomas, Highland, & Fromme, 2021). The mechanisms that mediate these Rab-switches are not completely understood. Specifically, in *Drosophila* larval salivary glands, Rab1 and Rab11 association to SGs appear to be crucial for SG maturation and timely exocytosis (Neuman et al., 2021). We found that the levels of Rab1 and Rab11 on SGs are respectively positively and negatively regulated by the exocyst, and that additionally, Rab11 is itself required for recruitment of Sec15 to immature SGs. These complex relationships between Rab11, Rab1 and the exocyst can be explained through a single-negative feedback loop that would guarantee adequate levels of Rab11 and the exocyst on maturing SGs. This possibility is supported by previous evidence from *Drosophila* and other systems that indicate that Rab11 and Sec15 interact physically and genetically (Escrevente et al., 2021; Guo et al., 1999; Novick, 2016). In fact, the exocyst is an effector of Rab11 (Sec4p in yeasts), as GTP-bound Rab11 recruits Sec15 to secretory vesicles, which in turn triggers formation of the holocomplex on the vesicle membrane as previously shown in yeast, *Drosophila* and mammalian cells (S. Wu et al., 2005) (Takahashi et al., 2012) (Zhang, Ellis, Sriratana, Mitchell, & Rowe, 2004).

An increasing number of publications reveal the complexity and variety of vesicle trafficking routes that feed into the biogenesis, maturation and exocytosis of SGs. Our study has revealed that the exocyst holocomplex acts at several steps of the pathway of regulated exocytosis: namely, maintaining Golgi complex morphology to allow protein exit from the ER and transit through the early secretory pathway; promoting homotypic fusion of SGs and acquisition of maturation factors; and finally, allowing SG-plasma membrane fusion. We propose that all these events depend on the activity of the exocyst complex as a tethering factor.

Materials and methods

Fly stocks and genetics

All fly stocks were kept on standard corn meal/agar medium at 25°C. Crosses were set up in vials containing two males and five females of the required genotypes at 25°C. Crosses were flipped every 24 hours to avoid larval overcrowding, and then moved to the desired temperature: 29°, 25°, 21°, 19° or 18°C. The temperature used for each experiment is specified in figure legends. In the experiments of **Figures 2**, **3**, **8** and **Supplementary Figures 1** and **11**, all crosses were maintained in a bath water to reduce temperature fluctuations. *D. melanogaster* lines used in this work are listed in **Supplementary Table 2**, and were obtained from the Bloomington Drosophila Stock Center (*http://flystocks.bio.indiana.edu*) or from the Vienna Drosophila Stock Center (*https://stockcenter.vdrc.at*); Sgs3-dsRed was generated by A.J. Andres' Lab (University of Nevada, United States).

Sgs3-GFP retention phenotype

Larvae or prepupae of the desired genotype were visualized and photographed inside glass vials under a fluorescence dissection microscope Olympus MVX10. In prepupae, localization of Sgs3-GFP or Sgs3-dsRed inside salivary glands or outside the puparium was determined. Each experiment was repeated at least three times.

Developmental staging and SG size

When larvae were cultured at 25°C, SG maturation progressed according to the timeline shown in **Figure 1**. Roughly, at 29°C larval development was shortened by 24 hours, and extended in 1, 3 and 4 days when larvae were cultured at 21, 19°C and 18 °C respectively. Precise physiological staging of each salivary gland was carried out according to Neuman and co-workers (Neuman et al., 2021) upon dissection and observation under the confocal microscope. In all experiments, only SGs from the distal-most cells of the salivary glands were imaged to avoid potential variations in SG size due to desynchronization in the synthesis of Sgs3. Data of temperatures and developmental staging from each experiment are synthesized in **Supplementary Table 4**.

RNA extraction and cDNA synthesis

Total RNA was isolated from dissected salivary glands of third-instar larvae using 500 µl of Quick-Zol reagent (Kalium Technologies, RA00201) following the manufacturer's instructions. The concentration and integrity of the RNA were determined using NanoDrop (Thermo Fisher Scientific) spectrophotometry. RNA (1 µg) was reverse-transcribed using M-MLV Reverse Transcriptase (Invitrogen, 10338842) using oligo-dT as a primer (*https://doi.org/10.1080/15548627*

.2021.1991191). Control reactions omitting reverse transcriptase were used to assess the absence of contaminating genomic DNA in the RNA samples. An additional control without RNA was included.

Real-time PCR

Gene expression was analyzed by quantitative PCR in a CFX96 Touch (Bio-Rad) cycler. The reactions were performed using HOT FIREPol EvaGreen qPCR Mix Plus (without ROX; Solis BioDyne, 08-25-00001), 0.40 μM primers, and 12–25 ng of cDNA, in a final volume of 10.4 μl (*https://doi.org/10.1080/15548627.2021.1991191*). Cycle conditions were initial denaturation at 95°C for 15 min, and 40 cycles of denaturation at 95°C for 20 s, annealing at 60°C for 1 min, and extension and optical reading stage at 72°C for 30 s, followed by a dissociation curve consisting of ramping the temperature from 65 to 95°C while continuously collecting fluorescence data (*https://doi.org/10.1080/15548627.2021.1991191*). Product purity was confirmed by agarose gel electrophoresis. Relative gene expression levels were calculated according to the comparative cycle threshold (CT) method (*https://doi.org/10.1080/15548627.2021.1991191*). Normalized target gene expression relative to rpl29 was obtained by calculating the difference in CT values, the relative change in target transcripts being computed as 2−ΔCT. The efficiencies of each target and housekeeping gene amplification were measured and shown to be approximately equal. Oligonucleotides were obtained from Macrogen (Seoul, Korea), and their sequences were the following: exo70: Fw 5'-GAAGTGGTTCTCCGATCGCT-3', Rv 5'-ACGAGCGGAGGTTGTCTTTT-3'; sec3: Fw 5'-GAAGACGCAACACATGGACG-3', Rv 5'-CTTTGCATATTGGCCCCATCC-3'; sec5: Fw 5'-GTCAATGAGACTGCCAAGAACT-3', Rv 5'-CCTGCAGTGGAATGTGCCTA-3'; rpl29: Fw 5'-GAACAAGAAGGCCCATCGTA-3', Rv 5'-AGTAAACAGGCTTTGGCTTGC-3'. Rpl29 was used as housekeeping gene. Specificity and quality of oligonucleotide sequences for exo70, sec3, sec5, and rpl29 were checked using Primer Blast Resource of the NCBI (*http://www.ncbi.nlm.nih.gov/tools/primer-blast/*).

Quantification of the penetrance of phenotypes upon knockdown of exocyst subunits

Sgs3-GFP intracellular distribution was analyzed in salivary gland cells, and one of three phenotypic categories was defined for each cell: 1) “mesh-like structure” when Sgs3-GFP was distributed in a network-like compartment; 2) “Immature SGs” when Sgs3-GFP was in SGs with a median diameter smaller than 3 μm; and 3) “Mature SGs” when Sgs3-GFP was in SGs with a median diameter equal or larger than 3 μm. The penetrance of each of the three phenotypes was calculated for each genotype of interest at the four different temperatures analyzed.

Salivary gland imaging

Salivary glands of the desired stage were dissected in cold PBS (137 mM NaCl, 2.7 mM KCl, 4.3 mM Na_2HPO_4, 1.47 mM KH_2PO_4, [pH 8]), and then imaged directly under the confocal microscope without fixation for no more than five minutes. In the experiment of **Figure 9G** and **Supplementary Figure 6E**, salivary glands were fixed for 2 hours in 4% PFA (Paraformaldehyde, Sigma) at room temperature, and then washed three times for 15 minutes with PBS-0.1% Triton X-100. For filamentous actin staining, salivary glands were incubated for 1 hour with Alexa Fluor 647 Phalloidin (ThermoFisher Scientific 1:400) in PBS-0.1% Triton X-100; stained tissues were mounted in gelvatol mounting medium (Sigma) and imaged at a confocal microscope Carl Zeiss LSM 710 with a Plan-Apochromat 63X/1.4NA oil objective, or Carl Zeiss LSM 880 with a Plan-Apochromat 20X/0.8 NA air objective or a Plan-Apochromat 63X/1.4NA oil objective.

For live imaging, salivary glands were dissected in PBS, and then mounted in a 15 mm diameter plastic chamber with a glass bottom made of a cover slip, containing 40 μl of HL3.1 medium (70 mM NaCl, 5 mM KCl, 1.5 mM CaCl2, 2 mM MgCl2, 5 mM HEPES, 115 mM sucrose, 5 mM trehalose, and pH 7.2 with NaHCO3). The medium was removed allowing the tissue to adhere to the bottom of the chamber, and then a Biopore membrane hydrophilic PTFE with a pore size of 0.4 µm (Millipore, Sigma) was placed over the sample, and HL3.1 medium was added on top covering the membrane. Images were captured under an inverted Carl Zeiss LSM 880 confocal microscope with a Plan-Apochromat 20X/0.8 NA air objective or a Plan-Apochromat 63X/1.4NA oil objective. For Supplementary Movie 1, frames were obtained every 2.25 seconds, while for Supplementary Movie 6 and 7, frames were captured every 0.78 and 0.67 seconds respectively.

Image processing and analysis

Image deconvolution was performed using the Parallel Spectral Deconvolution plugin of the ImageJ software (NIH, Bethesda, MD) with standard pre-sets (Schneider, Rasband, & Eliceiri, 2012). Image analyses were made with ImageJ (Schneider et al., 2012), and graphs were generated with the R Studio software (Team, 2020). For SG quantification, a region of interest (ROI) from each cell was used. In each ROI, the area of SGs was assessed, and SG diameter was calculated assuming that SGs are circular, using the formula $((\text{Area}/\pi)^{1/2})*2$ = Diameter. In experiments of **Figure 4B, D, F, H** and **Supplementary Figure 7A, C, E, G, I** two-dimensional lines scans were generated with the ImageJ plot profile. Fluorescence intensity was determined related to maximal intensity of each marker, always within a linear range. For quantification of fluorescence intensity of Syt1, CD63, Rab1 and Rab11 (**Figure 8** and **Supplementary Figure 11**), the mean intensity of three different ROIs of 5 μm^2 from each cell was measured. 17-35 cells from 4-11 salivary glands were used in the analysis. Association analyses in **Figure 5B, E** were performed considering structures as associated when the fluorescence spikes, of each fluorophore, had a distance minor or equal to ~0.6 µm. In the experiment of **Figure 7C**, GFP-Sec15 *foci* between granules were measured over total foci in four different ROIs of 225 μm^2 from each cell. For nucleus quantification (**Supplementary Figure 5E**) the area from each one, was measured using Image J freehand section. In polarity experiments, the mean fluorescence intensity of apical markers (**Supplementary Figure 6A, B**) was obtained by drawing a plotted line, of ~2 µm wide, over the apical membrane. In the case of PI(4,5)P_2 and F-actine (**Supplementary Figure 6 D, E**), localized in both compartments particularly enriched in apical membrane, a perpendicular line, of ~9 µm wide, was draw from the basolateral region to the apical and the coefficient between the maximum fluorescence intensity of both regions was calculated. Pearson's coefficient was obtained with JACoP plugging of Image J (Bolte & Cordelieres, 2006). The Golgi defect, upon exocyst knock down (**Supplementary Figure 9**), was measured as the number of cells with aberrant *cis*-Golgi or with swollen *trans*-Golgi vesicles over total cells within each salivary gland.

Statistical analyses

Statistical significance was calculated using one-way analysis of variance (ANOVA), a Likelihood Ratio test or a Wald Test, and followed by a Tuckey's test with a 95% confidence interval ($p < 0.05$) when comparing multiple factors.

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Additional information

Competing interests

The authors declare no competing or financial interests.

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Author contributions

SSF, Conceptualization, Data curation, Formal analysis, Validation, Investigation, Methodology, Writing – review and editing; SPP, Investigation, Methodology; SF, Investigation, Writing – review and editing; PW, Conceptualization, Funding acquisition, Project administration, Writing – review and editing; MM, Conceptualization, Supervision, Funding acquisition, Methodology, Writing – original draft, Project administration, Writing – review and editing

Supplementary tables and figures

Hours AEL	n	Cells quantified	SG quantified	Mean (µm)	Median (µm)	Minimum (µm)	Maximum (µm)
96-100	3	7	57	0.92	0.91	0.62	1.49
100-104	3	18	173	2.22	2.11	0.89	4.76
104-108	3	12	112	2.53	2.53	1.50	5.78
108-112	4	13	96	3.18	3.18	1.78	4.81
112-116	3	10	86	4.22	4.22	1.16	6.59
116-120	3	15	107	4.48	4.45	2.47	7.13

Supplementary Table 1.

Quantification of SG diameter at the indicated time intervals a=er egg laying.

SGs from salivary gland distal-most cells were analyzed. Columns display: Hours AEL: hours aAer egg laying; n: number of salivary glands analyzed; number of cells analyzed; number of SGs measured; mean diameter; median diameter; minimum diameter and maximum diameter.

Line	Number	Stock Center
Sgs3-GFP	5885	Bloomington
Sgs3-dsRed	-	A.J. Andres' Lab
YFP-Rab11	62549	Bloomington
YFP-Rab1	62539	Bloomington
UAS-dicer2	24650	Bloomington
UAS-White-RNAi	33613	Bloomington
UAS-Cherry-RNAi	35785	Bloomington
UAS-Rab11-RNAi	27730	Bloomington
UAS-Sec3-RNAi	35806	Vienna
UAS-Sec5-RNAi	27526	Bloomington
UAS-Sec5-RNAi	28873	Vienna
UAS-Sec6-RNAi	27314	Bloomington
UAS-Sec8-RNAi	45032	Vienna
UAS-Sec10-RNAi	27483	Bloomington
UAS-Sec15-RNAi	27499	Bloomington
UAS-Sec15-RNAi	35161	Vienna
UAS-Exo84-RNAi	108650	Vienna
UAS-Exo84-RNAi	28712	Bloomington
UAS-Exo70-RNAi	27867	Vienna
UAS-Exo70-RNAi	28041	Bloomington
UAS-PLCγ-PH-EGFP	58362	Bloomington
UAS-Synaptotagmin1-GFP	6925	Bloomington

Supplementary Table 2.

List of the *Drosophila* lines utilized in this work.

Stock number and repository center from where each line was obtained are indicated.

UAS-Sec8	9556	Bloomington
UAS-CD63-GFP	91390	Bloomington
UAS-Myr-Tomato	32221	Bloomington
UAS-GFP-KDEL	9898	Bloomington
UAS-Bip-SfGFP-HDEL	64749	Bloomington
UAS-RFP-Golgi	30908	Bloomington
UAS-GRASP65-GFP	8508	Bloomington
UAS-EGFP	5430	Bloomington
UAS-GFP-Sec15	39685	Bloomington
UAS-YFP-Rab11^{CA}	9791	Bloomington
tubP-GAL80[ts]	7017	Bloomington
UAS-mCD8-mCherry	27391	Bloomington
UAS-mito-GFP	8443	Bloomington
UASp-GFP-mCherry-Atg8a	37749	Bloomington

Supplementary Table 2. (continued)

Data Figure 3C						
Genotype	Temperature	Number of Glands Analyzed	Number of Distal Cells Analyzed	Phenotype	% of Phenotype	Standard Deviation
Control RNAi	29	4	12	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	25	5	13	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	21	5	17	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	19	5	15	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
Exo70 RNAi BL	29	11	43	Mesh Like Structure	12.73	28.67
				SG immature	87.27	28.67
				SG mature	0	0
	25	5	23	Mesh Like Structure	0	0
				SG immature	95.24	12.6
				SG mature	4.76	12.6
	21	7	23	Mesh Like Structure	0	0
				SG immature	90.48	25.2
				SG mature	9.52	25.2
	19	4	15	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
Sec3 RNAi	29	7	31	Mesh Like Structure	69.52	42.84
				SG immature	24.05	32.37
				SG mature	6.43	11.07
	25	7	20	Mesh Like Structure	16.67	28.87
				SG immature	78.57	28.41
				SG mature	4.76	12.6
	21	4	11	Mesh Like Structure	0	0
				SG immature	27.08	35.6
				SG mature	72.92	35.6
	19	9	32	Mesh Like Structure	0	0
				SG immature	33.52	33.75
				SG mature	66.48	33.75
Sec5 RNAi BL	29	4	14	Mesh Like Structure	96.43	7.14
				SG immature	3.57	7.14
				SG mature	0	0
	25	12	45	Mesh Like Structure	62.5	38.35
				SG immature	37.5	38.35
				SG mature	0	0
	21	9	20	Mesh Like Structure	27.78	44.1
				SG immature	72.22	44.1
				SG mature	0	0
	19	6	21	Mesh Like Structure	12.22	19.05
				SG immature	84.44	18.22
				SG mature	3.33	8.16
Sec10 RNAi	29	9	26	Mesh Like Structure	77.78	44.1
				SG immature	22.22	44.1
				SG mature	0	0
	25	6	34	Mesh Like Structure	59.17	46.95
				SG immature	40.83	46.95
				SG mature	0	0
	21	7	23	Mesh Like Structure	20.24	34.65
				SG immature	79.76	34.65
				SG mature	0	0
	19	6	20	Mesh Like Structure	0	0
				SG immature	100	0
				SG mature	0	0
Sec6 RNAi	29	9	34	Mesh Like Structure	66.67	50
				SG immature	33.33	50
				SG mature	0	0
	25	4	7	Mesh Like Structure	75	50
				SG immature	25	50
				SG mature	0	0
Sec8 RNAi	29	8	37	Mesh Like Structure	41.16	35.88
				SG immature	58.84	35.88
				SG mature	0	0
	25	6	20	Mesh Like Structure	8.33	20.41
				SG immature	80.56	30.58
				SG mature	11.11	27.22
	21	7	26	Mesh Like Structure	3.57	9.45
				SG immature	96.43	9.45
				SG mature	0	0
	19	6	22	Mesh Like Structure	0	0
				SG immature	96.67	8.16
				SG mature	3.33	8.16
Exo84 RNAi V	29	8	33	Mesh Like Structure	2.08	5.89
				SG immature	60.42	32.49
				SG mature	37.5	33.46
	25	4	14	Mesh Like Structure	0	0
				SG immature	27.92	28
				SG mature	72.08	28
	21	5	15	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	19	4	8	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
Sec15 RNAi V	29	4	17	Mesh Like Structure	100	0
				SG immature	0	0
				SG mature	0	0
	25	4	16	Mesh Like Structure	87.5	25
				SG immature	12.5	25
				SG mature	0	0
	21	6	25	Mesh Like Structure	43.33	46.33
				SG immature	56.67	46.33
				SG mature	0	0
	19	5	14	Mesh Like Structure	16.67	23.57
				SG immature	53.33	36.13
				SG mature	30	44.72

Data Supplementary Figure 1A						
Genotype	Temperature	Number of Glands Analyzed	Number of Distal Cells Analyzed	Phenotype	% of Phenotype	Standard Deviation
Control RNAi	29	4	12	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	25	5	13	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	21	5	17	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	19	5	15	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
Exo70 RNAi V	29	5	17	Mesh Like Structure	0	0
				SG immature	39.43	42.11
				SG mature	60.57	42.11
	25	6	15	Mesh Like Structure	0	0
				SG immature	8.33	20.41
				SG mature	91.67	20.41
	21	6	17	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
	19	4	15	Mesh Like Structure	0	0
				SG immature	0	0
				SG mature	100	0
Sec5 RNAi V	29	5	25	Mesh Like Structure	53.33	50.55
				SG immature	46.67	50.55
				SG mature	0	0
	25	5	12	Mesh Like Structure	20	27.39
				SG immature	70	27.39
				SG mature	10	22.36
	21	6	21	Mesh Like Structure	8.33	20.41
				SG immature	87.5	20.92
				SG mature	4.17	10.21
	19	6	27	Mesh Like Structure	5.56	13.61
				SG immature	91.11	14.4
				SG mature	3.33	8.16
Exo84 RNAi BL	29	7	38	Mesh Like Structure	93.2	12.85
				SG immature	6.8	12.85
				SG mature	0	0
	25	5	22	Mesh Like Structure	90	22.36
				SG immature	10	22.36
				SG mature	0	0
	21	6	27	Mesh Like Structure	95.83	10.21
				SG immature	4.17	10.21
				SG mature	0	0
	19	5	21	Mesh Like Structure	66.43	41.31
				SG immature	33.57	41.31
				SG mature	0	0
Sec15 RNAi BL	29	5	25	Mesh Like Structure	100	0
				SG immature	0	0
				SG mature	0	0
	25	6	26	Mesh Like Structure	94.44	13.61
				SG immature	5.56	13.61
				SG mature	0	0
	21	5	14	Mesh Like Structure	100	0
				SG immature	0	0
				SG mature	0	0
	19	5	20	Mesh Like Structure	100	0
				SG immature	0	0
				SG mature	0	0

Supplementary Table 3.

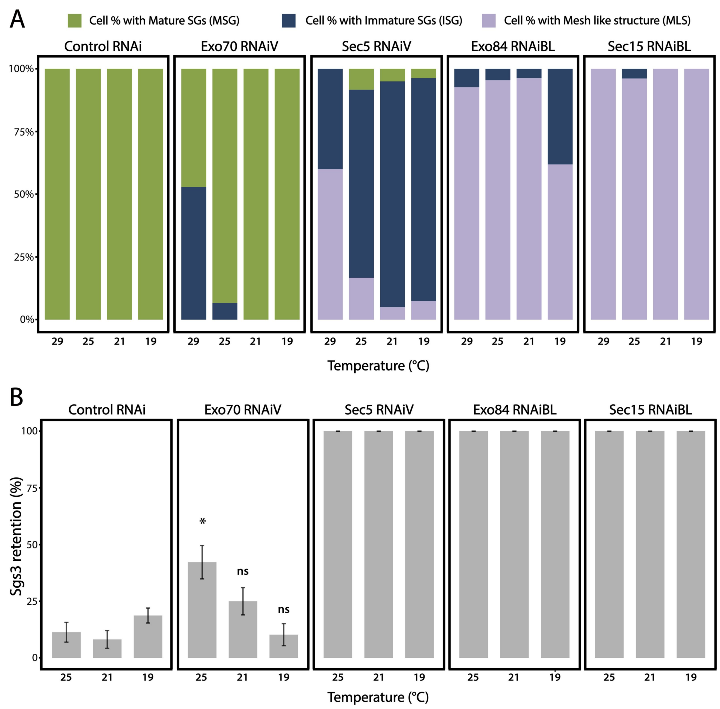
Raw data for experiments of Figure 3C and Supplementary Figure 1A.

Supplementary Table 4.

Salivary gland developmental staging.

Salivary glands analyzed in experiments of the indicated figures. Columns display: Experimental temperature, larval hours of development after egg laying (AEL), equivalent hours of development at 25°C (based on SG phenotype and salivary gland general appearance), presence or absence of Sgs3 in salivary glands, at the developmental time studied, and expected stage of SGs (Mature or Immature).

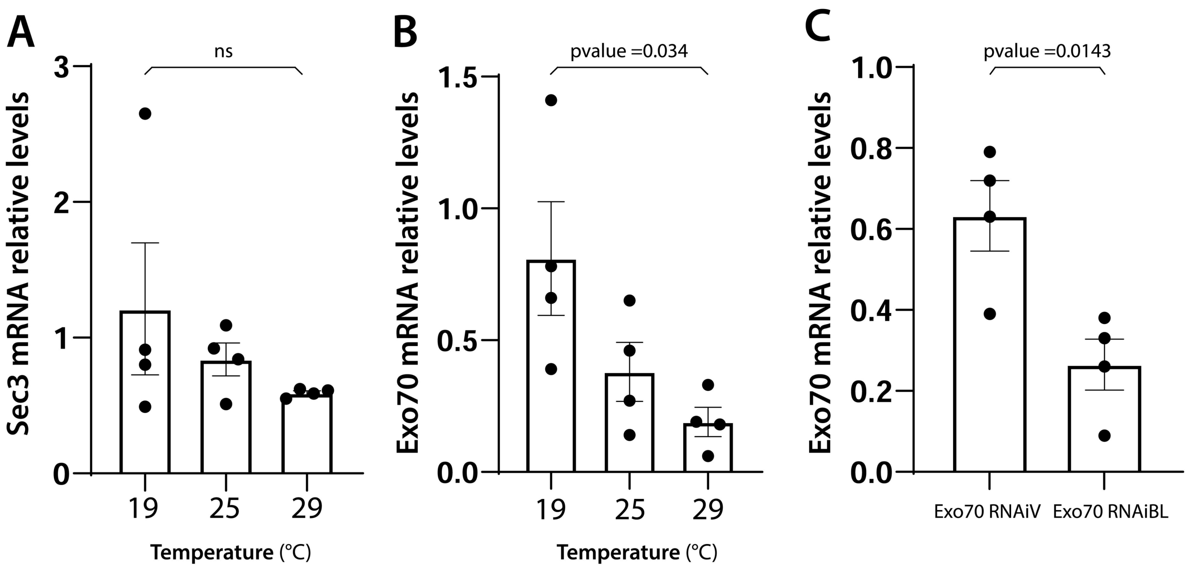
	Experimental temperature (°C)	Hours of development at the experimental temperature (hours AEL)	Equivalent hours of development at 25°C (hours AEL)	Presence of Sgs3	Expected developmental stage of SGs in wild type
Figure 2	29	~96	116-120	Yes	Mature
Figure 3	19, 21, 25, 29	~216, ~144, ~120, ~96	116-120	Yes	Mature
Figure 4	29	~96	116-120	Yes	Mature
Figure 5	18	~120-168	72-96	No	-
Figure 6	29	~96	116-120	Yes	Mature
Figure 7 A, E-H	29	~96	116-120	Yes	Mature
Figure 8 A, D, G, J	21	~108	96-104	Yes	Immature
Figure 8 B, E, H, K	21	~144	116-120	Yes	Mature
Figure 9 A-D	29	~96	116-120	Yes	Mature
Figure 9 E-G	18	~240	116-120	Yes	Mature
Supplementary Figure 1	29	~96	116-120	Yes	Mature
Supplementary Figure 2	19,25 or 29	~216, ~120, ~96	112-120	Yes	Mature
Supplementary Figure 3 A, D, G	29	~96	116-120	Yes	Mature
Supplementary Figure 3 B, E, H	120 hs at 18 °C and 36 hs at 29 °C	~156	116-120	Yes	Mature
Supplementary Figure 4	18	~240	116-120	Yes	Mature
Supplementary Figure 5 A-F	25	~72	~72	No	-
Supplementary Figure 6 A, E	25 or 29	~72 or ~50	~72	No	-
Supplementary Figure 6 B-D	25	~72	~72	No	-
Supplementary Figure 7	29	~96	116-120	Yes	Mature
Supplementary Figure 8	29	~96	116-120	Yes	Mature
Supplementary Figure 9	29	~96	116-120	Yes	Mature
Supplementary Figure 10 A, E	25	72-96	72-96	No	-
Supplementary Figure 10 B, F	25	100-104	100-104	Yes	Immature
Supplementary Figure 10 C, G	25	108-112	108-112	Yes	Mature
Supplementary Figure 10 D, H	25	116-120	116-120	Yes	Mature
Supplementary Figure 11 A, G, M	25 or 21	96-104 or ~108	96-104	Yes	Immature
Supplementary Figure 11 B, H, N	25 or 21	116-120 or ~144	116-120	Yes	Mature
Supplementary Figure 11 D, E, J, K, P, Q	25 or 21	116-120 or ~144	116-120	Yes	Mature
Supplementary Figure 12 A, B, E	29	~96	116-120	Yes	Mature
Supplementary Figure 12 D	29	~72	96-104	Yes	Immature



Supplementary Figure 1.

Penetrance of phenotypes observed after silencing subunits of the exocyst.

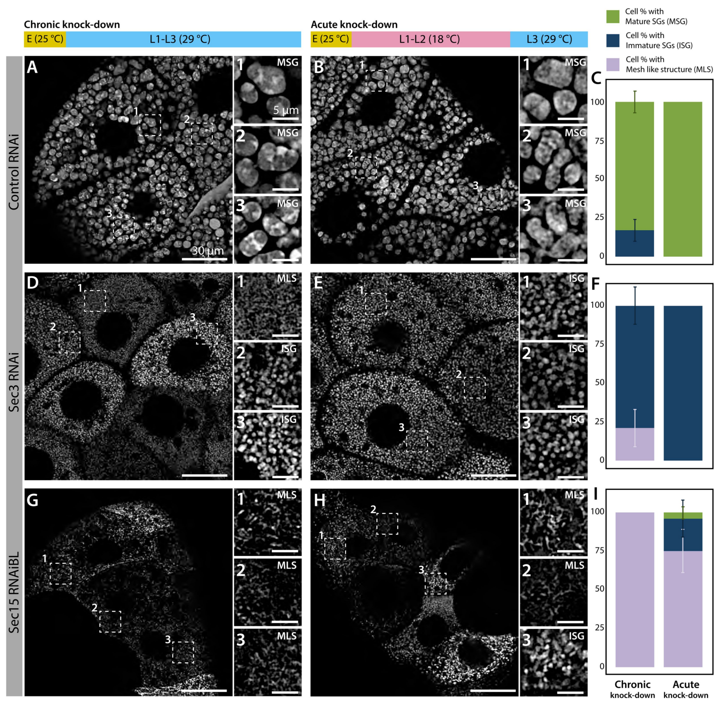
Additional RNA lines targeting exocyst subunits utilized to assess reduction-of-function phenotypes in the pathway of regulated exocytosis. (A) Quantification of the penetrance of each of the three phenotypes observed upon downregulation of each of the exocyst subunits. Larvae were grown at four different temperatures to achieve different levels of RNAi expression of exocyst subunits. “n” = Number of salivary glands analyzed. *control*^{RNAi}(*cherry*^{RNAi}) n = 4 (29 °C), n = 5 (25, 21 and 19 °C); *exo70*^{RNAiV} n = 5 (29 °C), n = 6 (25 °C), n = 6 (21 °C), n = 4 (19 °C); *sec15*^{RNAiBL} n = 5 (29 °C), n = 6 (25 °C), n = 5 (21 °C), n = 5 (19 °C); *exo84*^{RNAiBL} n = 7 (29 °C), n = 5 (25 °C), n = 6 (21 °C), n = 5 (19 °C); *sec5*^{RNAiV} n = 5 (29 °C), n = 5 (25 °C), n = 6 (21 °C), n = 6 (19 °C); (B) Quantification of the penetrance of the Sgs3-GFP retention phenotype in salivary glands of prepupae with the indicated RNAis at different temperatures. “n” = Number of vials containing 20-30 larvae each. *control*^{RNAi} (*cherry*^{RNAi}) n = 7 (25 °C), n = 9 (21 °C), n = 19 (19 °C); *exo70*^{RNAiV} n = 6 (25 °C), n = 8 (21 °C), n = 6 (19 °C); *sec5*^{RNAiV}, n = 4 (25 °C), n = 6 (21 °C), n = 5 (19 °C); *sec15*^{RNAiBL} n = 7 (25 °C), n = 8 (21 °C), n = 7 (19 °C); *exo84*^{RNAiBL} n = 5 (25 °C), n = 4 (21 °C), n = 8 (19 °C). RNAis were expressed using *fkh*-Gal4. Comparisons of the retention phenotype were carried out between genotypes at each different temperature, and statistical analysis was performed using a Likelihood ratio test followed by Tuckeýs test (“*” = p value < 0.05). For those RNAis with 100% penetrance no statistical analysis was performed. “ns” = not significant. This figure represents an extension of the experiments of **Figure 3C-D**; all RNAis were expressed with controls at the same time, and therefore the controls are identical to those of **Figure 3**.



Supplementary Figure 2.

Remaining mRNA levels after RNAi-mediated knock-down of exocyst subunits correlate with the observed phenotypes.

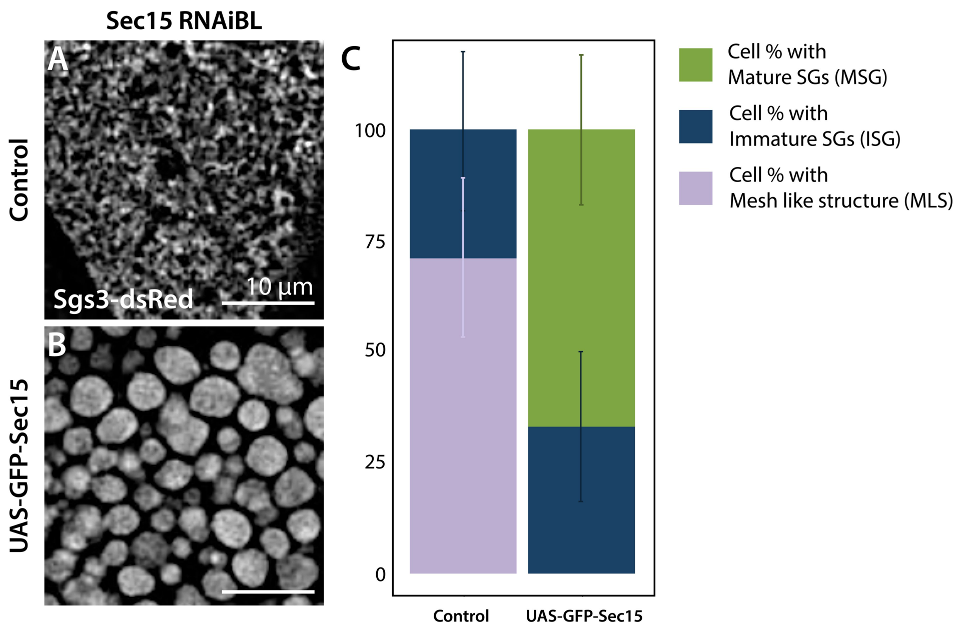
mRNA levels of the indicated genes were measured by qRT-PCR in homogenates from salivary glands. *sec3* (A) or *exo70* (B) mRNA levels from larva grown at the indicated temperatures were determined in salivary glands expressing *sec3*^{RNAi} (*sgs3-GFP*, *fkh-Gal4/ UAS-sec3*^{RNAi}) (A) or *exo70*^{RNAiBL}(*sgs3-GFP*, *fkh-Gal4/ UAS-exo70*^{RNAiBL}) (B) relative to salivary glands expressing *cherry*^{RNAi} (*sgs3-GFP*, *fkh-Gal4/ UAS-cherry*^{RNAi}) from larvae grown at the same temperatures. (C) *exo70* mRNA levels from larva grown at 29 °C were determined in salivary glands expressing *exo70*^{RNAiV} (*sgs3-GFP*, *fkh-Gal4/ UAS-exo70*^{RNAiV}) or *exo70*^{RNAiBL} (*sgs3-GFP*, *fkh-Gal4/ UAS-exo70*^{RNAiBL}) relative to salivary glands expressing *cherry*^{RNAi} (*sgs3-GFP*, *fkh-Gal4/ UAS-cherry*^{RNAi}). n = 4 for all genotypes and conditions. Statistical analysis was performed by one-way ANOVA followed by Tukey's test with a confidence interval higher than 95% ($p < 0.05$). "ns"= not significant.



Supplementary Figure 3.

Chronic or Acute knock-down of exocyst subunits generate comparable phenotypes.

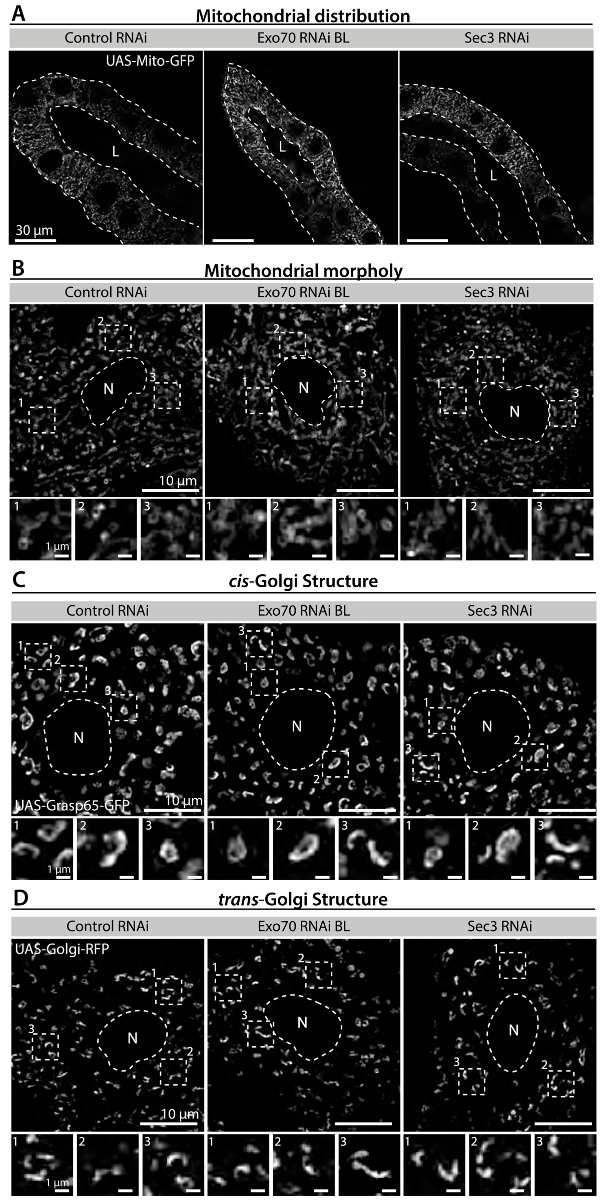
Phenotypes of control (*white*^{RNAi}) (A-C), *sec3*^{RNAi} (D-F) or *sec15*^{RNAiBL} (G-I) salivary glands. The thermo-sensitive Gal80 system (*Gal80*^{ts}) was utilized to compare salivary glands after silencing the indicated genes throughout larval development (chronic silencing: A, D, G) or at the 3rd larval instar only (acute silencing: B, E, H). Chronic RNAi expression was achieved by growing larvae at the restrictive temperature (29 °C) from larval eclosion onwards (A, D and G). Acute RNAi expression was achieved by growing larvae at the permissive temperature (18 °C) from larval eclosion until early L3 (larval 3rd instar), (B, E and H) and then shifting larvae to the restrictive temperature (29 °C) until analyzed. E: embryogenesis; L1: larval 1st instar; L2: 2nd instar; and L3: 3rd instar. The penetrance of each phenotype observed (Mature Secretory Granules, Immature Secretory Granules or Mesh-Like Structure) was quantified for each genotype and experimental condition (C, F and I). No substantial differences were detected between chronic and acute silencing of exocyst subunits. “n” represents the total number of salivary glands analyzed. *control*^{RNAi} (*white*^{RNAi}) (A) n=5, (B) n= 3; *sec3*^{RNAi} (D) n=6, (E)n=7; *sec15*^{RNAiBL}(G, H) n= 4.



Supplementary Figure 4.

Expression of GFP-Sec15 can rescue SG maturation after sec15 knock-down.

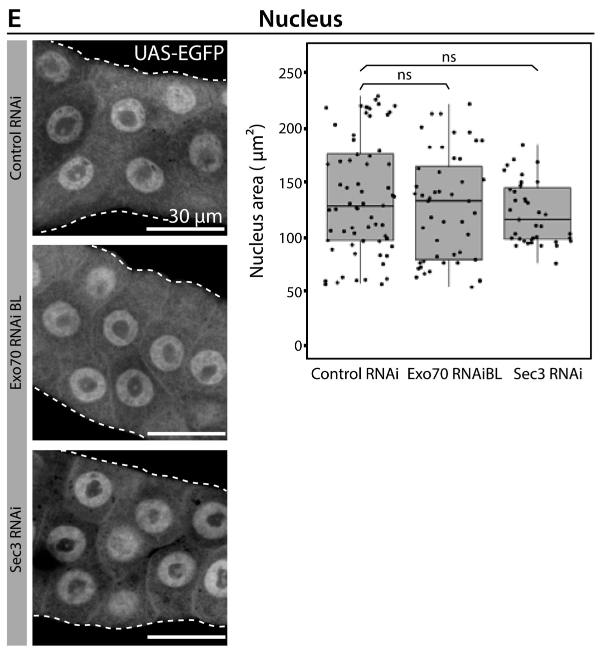
Expression of *sec15*^{RNAiBL}at 18 °C (*sgs3-dsRed; UAS-PLCγPH-GFP, fkh-Gal4/UAS-sec15*^{RNAiBL}*)* (A) generates salivary glands that do not form mature SGs (MSG) in any of the cells analyzed. Simultaneous expression of GFP-Sec15 (*UAS-GFP-sec15/ Sgs3-dsRed*; *UAS-sec15*^{RNAiBL}*/ fkh-Gal4)* (B) generates salivary glands with MSGs in 67% of the cells. (C) Quantification of each of the three possible phenotypes Mature SGs (MSG), Immature SGs (ISG) or Mesh-like structure (MLS) in control larvae (*sgs3-dsRed; UAS-PLCγPH-GFP, fkh-Gal4/UAS-sec15*^{RNAiBL}*)* or after expression of the rescue construct (*UAS-GFP-sec15/ sgs3-dsRed*; *UAS-sec15*^{RNAiBL}*/ fkh-Gal4).* "n" represents the total number of salivary glands analyzed. Control genotype (*UAS-PLCγPH-GFP*) n=7; Rescue genotype (*UAS-GFP-sec15*) n=5.

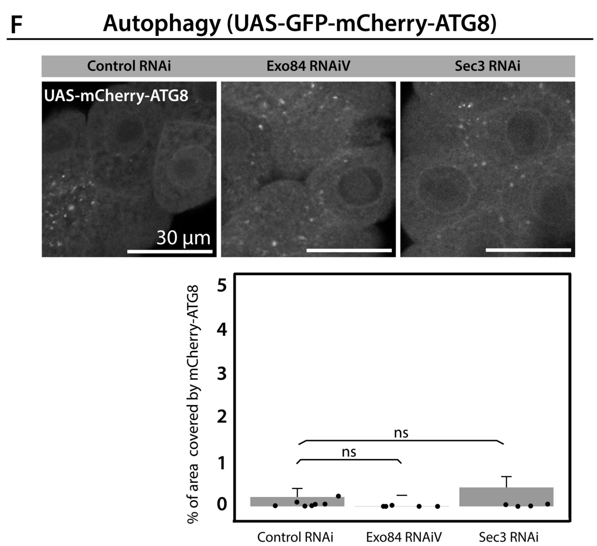


Supplementary Figure 5.

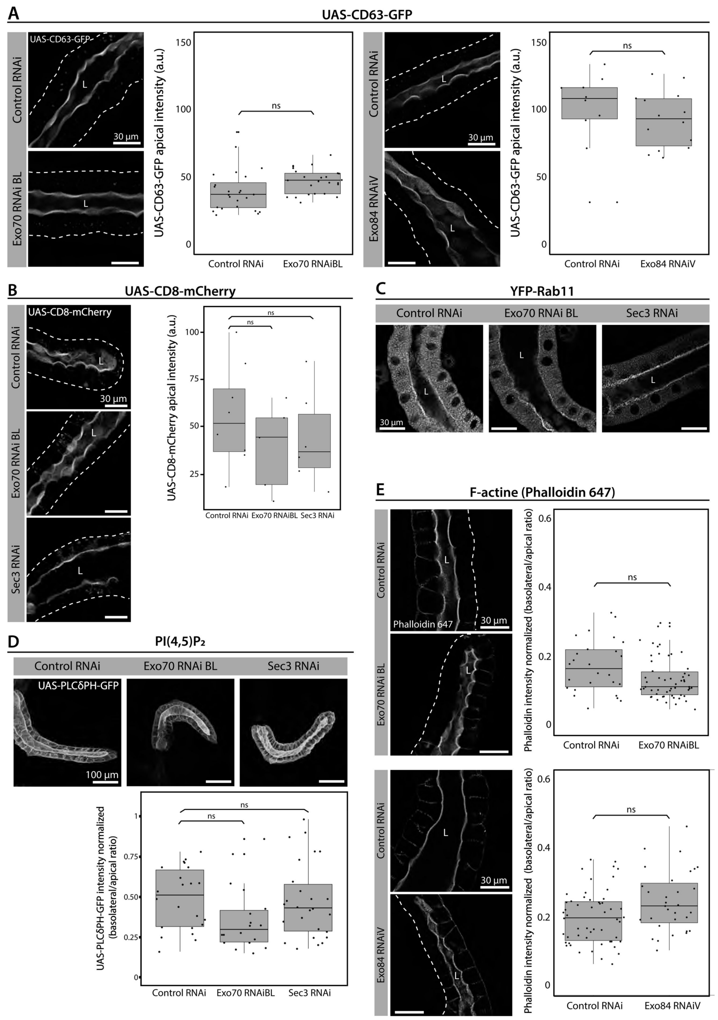
Exocyst down-regulation does not affect general cellular health or homeostasis.

Mitochondrial distribution and morphology (A-B), *cis*- and *trans*-Golgi complex distribution and morphology (C-D), nuclear size (E), and autophagy (F) were analyzed in salivary glands of control (*UAS-white*^{RNAi}) and exocyst knock-down salivary glands. Mitochondria were labelled with mito-GFP (*UAS-mito-GFP*), *cis*-Golgi with GRASP65 (*UAS-GRASP65-GFP), trans*-Golgi with RFP-Golgi (*UAS-RFP-Golgi*), nuclei with EGFP (*UAS-EGFP*), and autophagy with mChAtg8 (*UAS-GFP-mCh-atg8*). Larvae were grown at 25 °C to achieve a condition at which knock-down of the exocyst subunits mostly generate cells with immature SGs. RNAis were expressed using *fkh-Gal4*. (E) Quantification of nuclear area in the indicated genotypes was performed using image J. (F) Quantification of the area coved by mCh-Atg8 *foci* over the total area of salivary gland analyzed (%) was performed using image J. Statistical analysis was carried-out using one-way analysis of variance (ANOVA) "n" represents the total number of salivary glands analyzed. (E) *control*^{RNAi} (*white*^{RNAi}) n= 8; *exo70*^{RNAiBL} n=5; *sec3*^{RNAi} n=4. (F) starved *control*^{RNAi}(*white*^{RNAi}) n= 5; fed *control*^{RNAi} (*white*^{RNAi}) n= 8; fed *exo84*^{RNAiV} n=5; fed *sec3*^{RNAi} n=5.





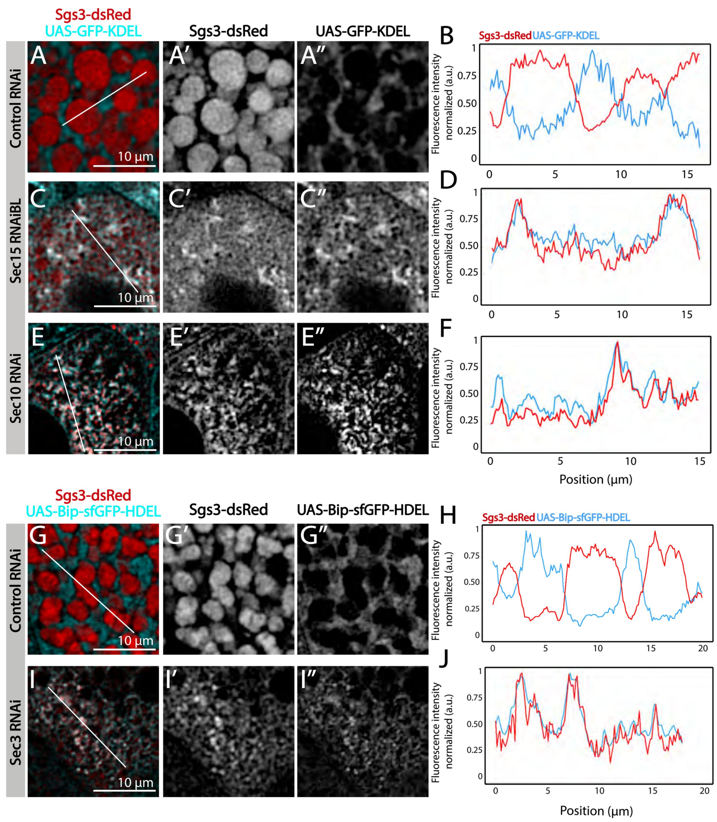
Supplementary Figure 5. (continued)



Supplementary Figure 6.

Exocyst down-regulation does not affect apical polarity markers.

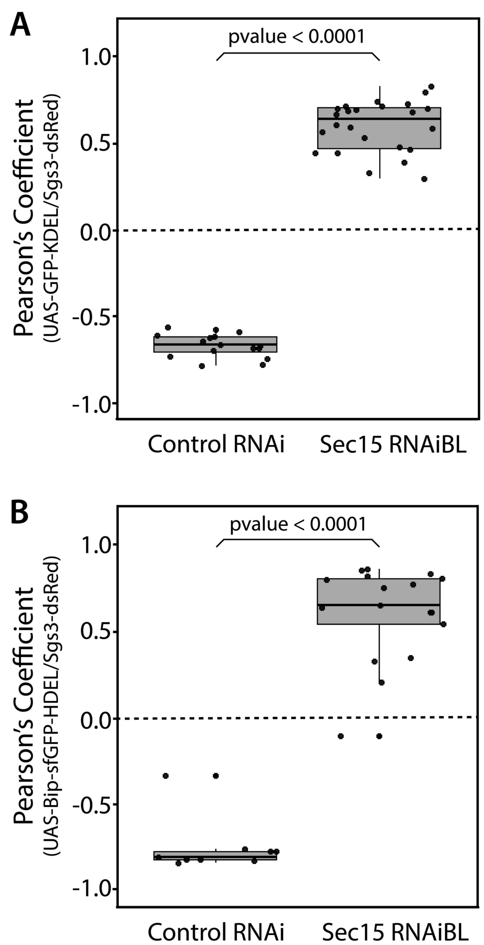
(A) CD63 (*UAS-CD63-GFP*), (B) CD8 (*UAS-CD8-mCherry*), (C) Rab11 (*YFP-Rab11*), (D) PI(4,5)P_2 (*UAS-PLC*γ*-PH-GFP)* and (E) filamentous actin (phalloidine-647) distribution were analyzed in salivary glands of control (*UAS-white^{RNAi}*) or exocyst knock-down salivary glands (*UAS-exo70^{RNAiBL}, UAS-sec3^{RNAi}* or *UAS-exo84^{RNAiV}*). Larvae were grown at 25 °C (*UAS-exo70^{RNAiBL} or UAS-sec3^{RNAi}*) or at 29°C (*UAS-exo84^{RNAiV}*) to achieve a condition at which exocyst downregulation generate mostly cells with immature SGs. Note that in (A) and (E), for accurate comparisons, the control genotype (*UAS-white^{RNAi}*) was also assayed at different temperatures. Transgenes were expressed with *fkh-Gal4*. Statistical analysis was performed using (A-B) one-way analysis of variance (ANOVA), (D) Likelihood ratio test followed by Tuckeýs test or (E) Wald Test (p value < 0.05, “ns”=not significant). “n” represents the total number of salivary glands analyzed. (A) *control^{RNAi}*(*white^{RNAi}*) n= 6; *exo70^{RNAiBL}* n=5; *control^{RNAi}*(*white^{RNAi}*) n=3; *exo84^{RNAiV}* n=3. (B) *control^{RNAi}*(*white^{RNAi}*) n= 8; *exo70^{RNAiBL}* n=5; *sec3^{RNAi}* n=6. (D) n=4. (E) *control^{RNAi}* (*white^{RNAi}*) n= 4; *exo70^{RNAiBL}* n=10; *control^{RNAi}*(*white^{RNAi}*) n=6; *exo84^{RNAiV}* n=4.



Supplementary Figure 7.

The exocyst is required for Sgs3-GFP exit from ER and SG biogenesis.

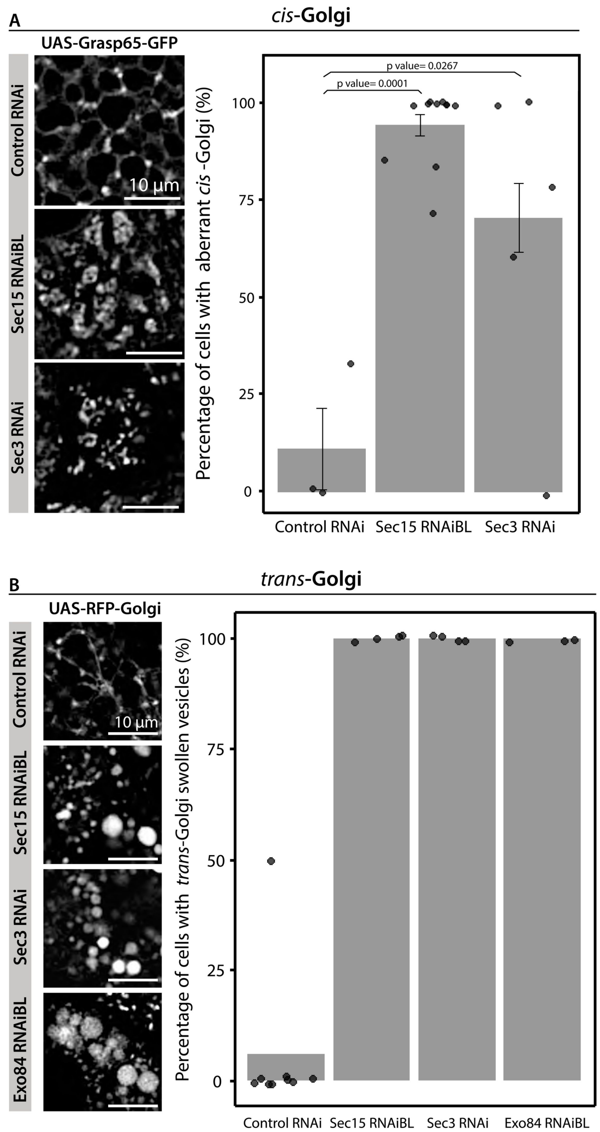
Confocal images of unfixed salivary glands of the indicated genotypes. Larvae were grown at 29 °C to achieve maximal activation of the Gal4-UAS system, and maximal downregulation of exocyst complex subunits. The ER was labeled with GFP-KDEL (*UAS-GFP-KDEL*), (A, C and E) or with Bip-sfGFP-HDEL (*UAS-Bip-sfGFP-HDEL*) (G and I). In control salivary glands (*fkh-Gal4/UAS-white^{RNAi}*) (A and G), the ER displayed a network-like appearance localized in between mature SGs that contain Sgs3-dsRed, and thus no colocalization between the ER marker and Sgs3-dsRed was detected (B and H). Upon expression of *sec15^{RNAi}* (*fkh-Gal4/UAS-sec15^{RNAiBL}*) (C), sec10^{RNAi} (*UAS-sec10^{RNAi}/ fkh-Gal4*) (E) or *sec3^{RNAi}* (*UAS-sec3^{RNAi}; fkh-Gal4*) SGs did not form, and Sgs3-dsRed was retained in the ER, as demonstrated by colocalization of both markers in the two-dimensional line scan analysis (D, F and J). RNAis were expressed using *fkh*-Gal4. Scale bar 10µm.



Supplementary Figure 8.

Quantification of colocalization between ER markers and Sgs3.

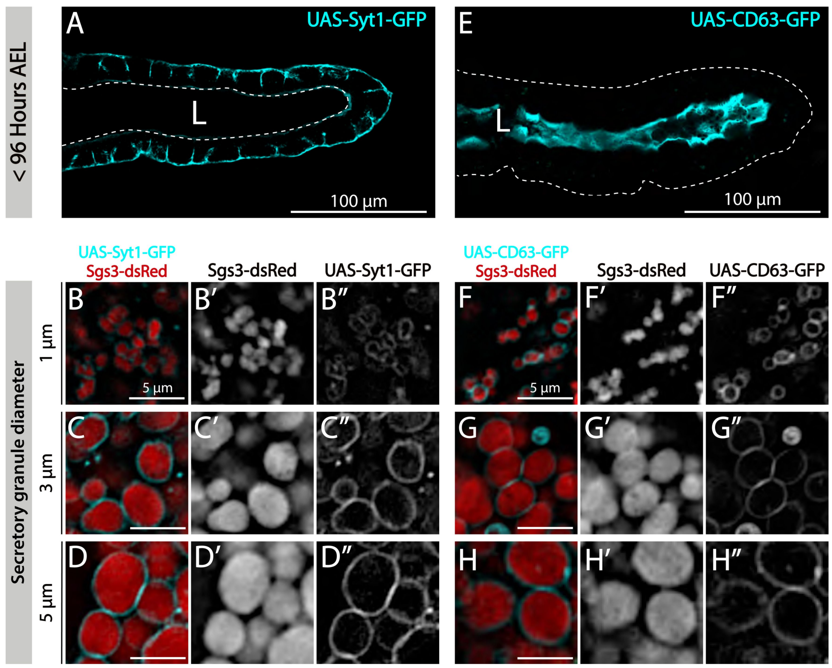
Pearsońs coefficient to evaluate colocalization between the ER markers *UAS-GFP-KDEL* (A) or *UAS-Bip-sfGFP-HDEL* (B) and Sgs3-dsRed in control salivary glands (*UAS-white^{RNAi}*) or after exocyst knock-down (*UAS-sec15^{RNAiBL}*). Larvae were grown at 29 °C. Statistical analysis was performed using one-way analysis of variance (ANOVA). Analysis corresponds to data shown in **Figure 4**. "n" represents the total number of salivary glands analyzed. (A) n=3, (B) n=2.



Supplementary Figure 9.

The exocyst is required to maintain the typical morphology of the Golgi complex.

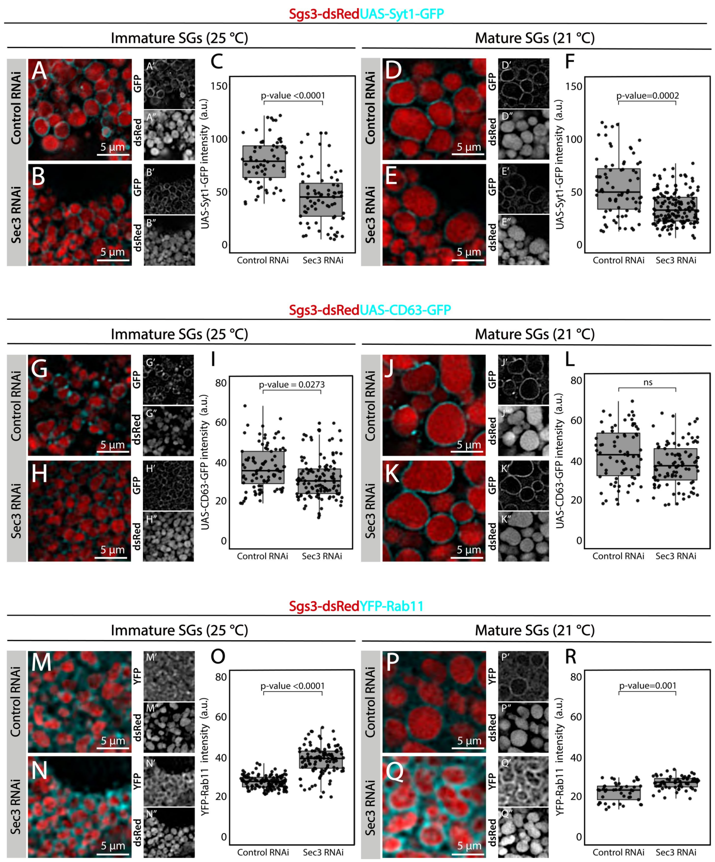
Confocal images of unfixed salivary glands of the indicated genotypes. Larvae were grown at 29 °C to achieve maximal activation of the Gal4-UAS system, and maximal downregulation of exocyst subunits. *cis*-Golgi (*UAS-Grasp65-GFP*) (A) and trans-Golgi (*UAS-RFP-Golgi*) (B) markers were analyzed In control salivary glands (*fkh-Gal4/ UAS-white^{RNAi}*) or in exocyst-down-regulated salivary glands (UAS-sec15^{RNAiBL}, UAS-sec3^{RNAi} or UAS-exo84^{RNAiV}). RNAis were expressed using *fkh*-Gal4. Scale bar 10 µm. Bar graphs show the quantification of the penetrance of the Golgi complex phenotype. Statistical analysis was performed using a Likelihood ratio test followed by Tuckeýs test (p value < 0.05, “ns”=not significant). “n” represents the total number of salivary glands analyzed. (A) *control^{RNAi}*(*white^{RNAi}*) n= 3; *sec15^{RNAiBL}* n=11; *sec3^{RNAi}*n=3. (B) *control^{RNAi}* (*white^{RNAi}*) n= 9; *sec15^{RNAiBL}* n=4; *sec3^{RNAi}* n=4; *exo84^{RNAiV}* n=3.



Supplementary Figure 10.

Incorporation of maturation markers to SGs in wild type salivary glands.

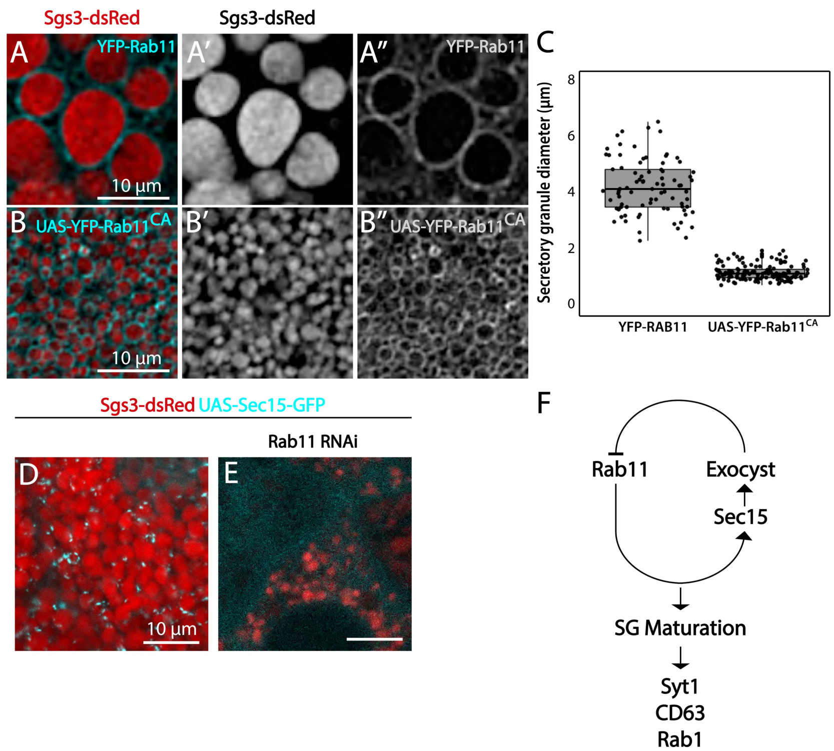
Incorporation of the SG maturation markers Syt1 (*sgs3-dsRed/ UAS-syt1-GFP; fkh-Gal4*), (A-D) and CD63 (*sgs3-dsRed/ UAS-CD63-GFP; fkh-Gal4*), (E-H) to developing granules. Prior to SG biogenesis (<96 hours AEL) Syt1 localized at the basolateral plasma membrane of salivary cells (A), while CD63 localized at the apical plasma membrane (E). Scale bar 100 µm. Both markers localized at the membrane of immature SGs and mature SGs (B-D and F-H). Scale bar 5 µm. Note that Syt1 signal intensity increases around SGs as they mature (compare B", C" and D"), whereas CD63 signal remains fairly stable during SG maturation (compare F", G" and H").



Supplementary Figure 11.

The exocyst mediates the acquisition of secretory granule maturation factors.

Maturation proteins Syt-1 and CD63 were gradually incorporated to the membrane of SG. Confocal images of unfixed salivary glands of the indicated genotypes. Larvae were grown at 25 or 21 °C to modulate the activity of the Gal4-UAS system and generate a maximal proportion of immature (A, B, G, H, M and N) or mature SGs (D, E, J, K, P and Q) respectively. (A, B, D and E) Recruitment of Syt1-GFP (*UAS-Syt1-GFP*); of (G, H, J and K) CD63-GFP (*UAS-CD63-GFP*); or of (M, N, P and Q) YFP-Rab11 was analyzed in control salivary glands expressing $white^{RNAi}$(*fkh-Gal4/ UAS-white*^{RNAi}) and in salivary glands expressing $sec3^{RNAi}$ (*UAS-sec3*^{RNAi}*; fkh-Gal4*). SGs were labelled with Sgs3-dsRed. Fluorescent intensity around SGs of each of the analyzed maturation factors was quantified using the ImageJ software and plotted (C, F, I, L, O and R). Comparison of fluorescence intensity among genotypes, and statistical analysis were performed using one-way analysis of variance (ANOVA). "n" represents the number of salivary glands: (C) $control^{RNAi}$ n= 7, $sec3^{RNAi}$n=6; (F) $control^{RNAi}$ n=7, $sec3^{RNAi}$ n=13; (I) $control^{RNAi}$ n= 12, $sec3^{RNAi}$ n= 9; (L) $control^{RNAi}$ n= 8, $sec3^{RNAi}$ n= 8; (O) $control^{RNAi}$ n= 7, $sec3^{RNAi}$ n= 10; (R) $control^{RNAi}$ n= 7, $sec3^{RNAi}$ n= 4. Transgenes were expressed using *fkh-Gal4*. "ns" = not significant. Scale bar 5 μm.



Supplementary Figure 12.

The exocyst negatively regulates incorporation of Rab11 to SGs.

Salivary gland cells expressing Sgs3-dsRed and endogenously tagged YFP-Rab11 (A), or overexpressing a constitutively active form of Rab11 at 29 °C (*fkh-Gal4/ UAS-YFP-Rab11^{CA}*) (B). YFP-Rab11 localized around mature SGs (A). Overexpression of Rab11^{CA} arrested SG maturation (B), as shown in the quantification of SG diameter (C). YFP-Rab11, n = 6; UAS-YFP-RAB11^{CA}, n = 5. "n" = Number of salivary glands. (D) In control salivary glands, GFP-Sec15 localized as *foci* on the periphery of mature SGs (*sgs3-dsRed/ UAS-GFP-sec15; fkh-Gal4*). (E) Knock-down of Rab11 (*sgs3-dsRed; UAS-rab11^{RNAi}/ fkh-Gal4*) generated immature SGs and prevented GFP-Sec15 from localizing on SGs. (F) Proposed mechanism of genetic interactions between Rab11 and Sec15. A single-negative feedback loop regulates the levels of these proteins on the SGs, and recruitment of the maturation factors Syt-1, CD63 and Rab1. Scale bar 10 µm.

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Editors

Reviewing Editor
Benjamin Glick
The University of Chicago, Chicago, United States of America

Senior Editor
Sofia Araújo
University of Barcelona, Barcelona, Spain

Reviewer #1 (Public review):

Suarez-Freire et al. analyzed here the function of the exocyst complex in the secretion of the glue proteins by the salivary glands of the Drosophila larva. This is a widely used, genetically accessible system in which the formation, maturation and precisely timed exocytosis of the glue secretory granules can be beautifully imaged. Using RNAi, the authors show that all units of the exocyst complex are required for exocytosis. They show that not just granule fusion with the plasma membrane is affected (canonical role), but also, with different penetrance, that glue protein is retained in the ER, secretory granules fail to fuse homotypically or fail to acquire maturation features. The authors document these phenotypes and postulate specific roles for the exocyst in these additional processes to explain them: exocyst as a Golgi-Golgi, Golgi-granule or granule-granule tether.

Compared to the initial submission, this revised version of the study presents strengthened evidence for these novel roles. In particular, authors show juxta-Golgi localization of exocyst components and disruption of the trans-Golgi compartment upon exocyst loss. Additionally, the revised study contains controls indicating that glue secretion defects prior to plasma membrane exocytosis are not due to polarity loss or unspecific poor health of cells.

https://doi.org/10.7554/eLife.92404.2.sa3

Reviewer #2 (Public review):

The manuscript from Wappner and Melani labs claims a novel for the exocyst subunits in multiple aspects of secretory granule exocytosis. This an intriguing paper for it suggests multiple roles of the exocyst in granule maturation and fusion with roles at the ER/Golgi interface, TGN, granule homotypic fusion.

A key strength is the breadth of the assays and study of all 8 exocyst subunits in a powerful model system (fly larvae). But why do KD of different exocysts have different effects on presumed granule formation? Also it can be hard to disentangle direct vs. secondary effects, as much of the TGN seems to be altered in the KDs. The authors ascribe many of the results to the holocomplex, but there are major differences between the proteins -- this may be all related to the different levels of expression (as the authors propose), but only limited mRNA was examined.

Unresolved Comments:

(A) Explanation variability of exocyst KD on the appearance of MSG. What is remarkable is a highly variable effect of different subunit KD on the percentage of cells with MLS (Fig. 4C). Controls = 100 %, Exo70=~75% (at 19 deg), Sec3 = ~30%, Sec10 = 0%, Exo84 = 100% ... This is interesting for the functional exocyst is an octameric holocomples, thus why the huge subunit variability in the phenotypes? One explanation is that the levels of KD varied between the subunits. Another is that not all subunits have equivalent roles (as seen for instance in exocyst's roles in autophagy).

This should be addressed by quantification of the KD of the 8 different exocyst proteins (and or mRNA as only 2 subunits were studied). If their data holds up then the underlying mechanism here needs to be considered. (Note: there is some precedent from the autophagy field of differential exocyst effects).

(B) Golgi: It is unclear from their model (Fig. 5) why after exocyst KD of Sec15 the cis-Golgi is more preserved than the TGN, which appears as large vacuoles.

(C) Granule homotypic fusion. Over-expression of just one subunit, Sec15-GFP, made giant secretory granules (SG) that were over 8 microns big. Does it act like a seed to promote exocyst assembly as the authors propose? If so is there evidence that there is biochemically more holocomplex with expression of Sec15, but not other subunits?

(D) The authors should better frame their interpretations of other studies of the exocyst that includes role in autophagy, Palade body trafficking and differential roles of the subunits.

In summary, there clearly are striking new effects on secretory granule biogenesis by dysfunction of the exocyst which are important and should inspire other studies for new roles of the exocyst; e.g. in non cannonical roles. Secondly, the power of the system to partially deplete proteins (if further validated) suggests that one may need to consider protein expression as an important variable that can be used to unmask multiple phenotypes in granule maturation. Last this paper implies new roles of the exocyst in homotypic fusion, which could be investigated in future work.

https://doi.org/10.7554/eLife.92404.2.sa2

Reviewer #3 (Public review):

Freire and co-authors examine the role of the exocyst complex during the formation and secretion of mucins from secretory granules in the larval salivary gland of Drosophila melanogaster. Using transgenic lines with a tagged Sgs3 mucin, the authors KD expression of exocyst subunit members and observe a defect in secretory granules with a heterogeneity of phenotypes. By carefully controlling RNAi expression using a Gal4-based system, the authors can KD exocyst subunit expression to varying degrees. The authors find that the stronger the inhibition of expression of the exocyst is, the earlier the defect is in the secretory pathway. The manuscript is well written, the model system is physiological, and the techniques are innovative.

In my initial review, my major concern was the pleiotropic effect of the loss of exocyst. The authors have responded to this point with clarity and have argued that the multiple localisations of exocyst during the Sgs3 synthesis programme indicate it is likely a direct phenotype. They also performed some analysis of PM lipids but did not detect a difference. I accept the arguments presented. However, I remain concerned that these are due to a pleiotropic effect. It is very hard to absolutely prove a direct effect, and due to the unusual claim and nature of the evidence (depletion levels), I think that there is still the possibility of this being an indirect effect. Perhaps it is just worth the authors writing a paragraph in the discussion, at least accepting the possibility that it is an indirect effect so future readers are aware of that.

https://doi.org/10.7554/eLife.92404.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***

> ***Reviewer 1:***
>
> *(1) General comment: The evidence for these highly novel, potentially interesting roles (of the exocyst) would need to be more compelling to support direct involvement.*

We wish to thank the reviewer for his/her comments, and for considering that the proposed functions are highly novel and potentially interesting. To strengthen the evidence supporting the new roles of the exocyst, we have performed a number of additional experiments that are depicted in novel figures or figure panels of the new version of the manuscript. Particularly, we aimed at providing further support of the direct involvement of the exocyst in different steps of the regulated secretory pathway. Please see the details below.

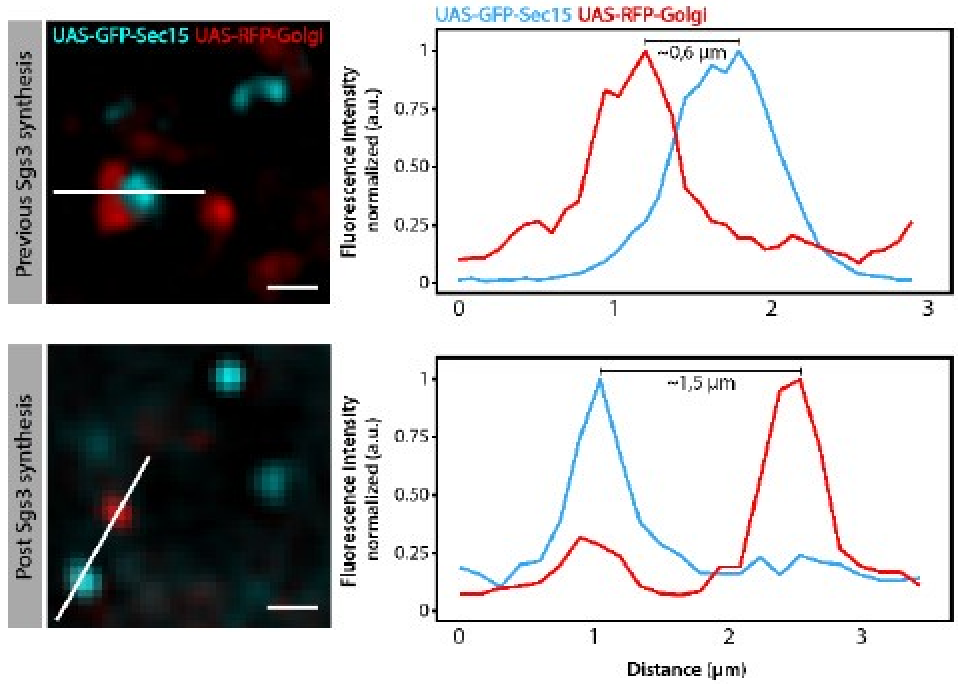
> *(2) For instance, the localization of exocyst to Golgi or to granule-granule contact sites does not seem substantial.*

We have performed quantitative colocalization studies, as suggested by the reviewer to further substantiate our initial findings. We have carefully analysed GFP-Sec15 distribution in relation to the Golgi complex and secretory Glue granules at relevant time points of salivary gland development. Overall, we found that GFP-Sec15 distribution is dynamic during salivary gland development. Before Glue synthesis (72 h AEL), Sec15 was observed in close association (defined as a distance equal to, or less than 0.6 μm) with the Golgi complex (please see below Author response image 1). This association was lost once Glue granules have begun to form (96 h AEL). Importantly, we do not see relevant association between GFP-Sec15 and the ER (please see Author response image 2). These observations support our conclusion that the exocyst plays a role at the Golgi complex. New images supporting these conclusions, as well as quantitative data, have been included in Figure 5 of the new version of the manuscript. In addition, real time imaging, as well as 3D reconstruction analyses, confirming the close association between Sec15 and Golgi cisternae are now included in the manuscript. Please see Supplementary Videos 1-3. These new data are described in the text lines 200-210 of the Results section and text lines 359368 of the Discussion section.

Interestingly, at the time when Sec15-Golgi association is lost (96 h AEL), Sec15 foci associate instead with newly formed secretory granules (< 1μm diameter). This association persists during secretory granule maturation (100-116 h AEL), when Sec15 foci localize specifically in between neighbouring, immature secretory granules. When maturation has ended and Glue granule exocytosis begins (116-120 h AEL), this localization between granules is lost. These observations are consistent with a role of the exocyst in homotypic fusion during SG maturation. We have included new images showing that association between Sec15 and secretory granules is dynamic and depends on the developmental stage. We have quantified this association both during maturation and at a stage when SGs are already mature. We have in addition performed a 3D reconstruction analysis of these images to confirm the close association between Sec15 and immature SGs. These new data are now depicted in Figure 7BC, Supplementary Videos 4-5, and described in text lines 216-221 of the Results section. In addition, a lower magnification image is provided below in this letter (Author response image 3), quantifying the proportion of Sec15 foci localized in between SGs (yellow arrows) relative to the total number of Sec15 foci (yellow arrows + green arrowheads).

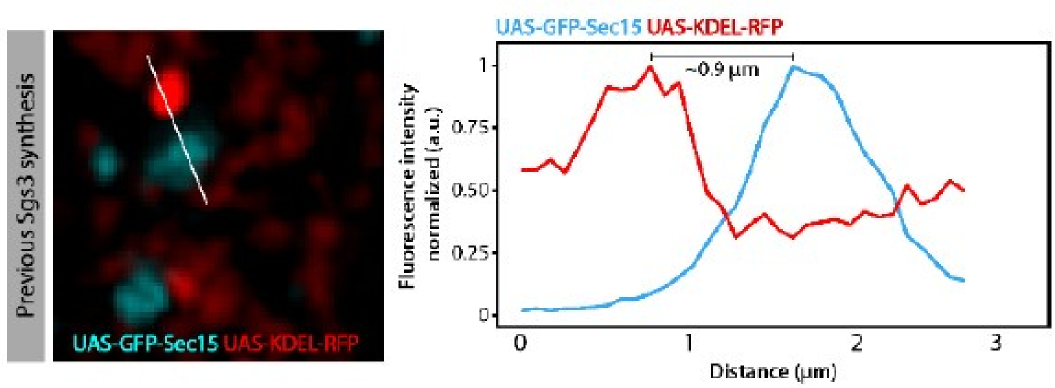
Author response image 1.

Criteria utilized to define Sec15 focithat were“associated” or“not associated” withthe trans-Golgi network in the experiments of Figure 5C-E of the manuscript.When the distance between maximal intensities of GFP-Sec15 and Golgi-RFP signals was equal or less than 0.6 m, the signals were considered “associated” (upper panels). When the distance was more than 0.6 m, the signals were considered “not associated” (lower panels).



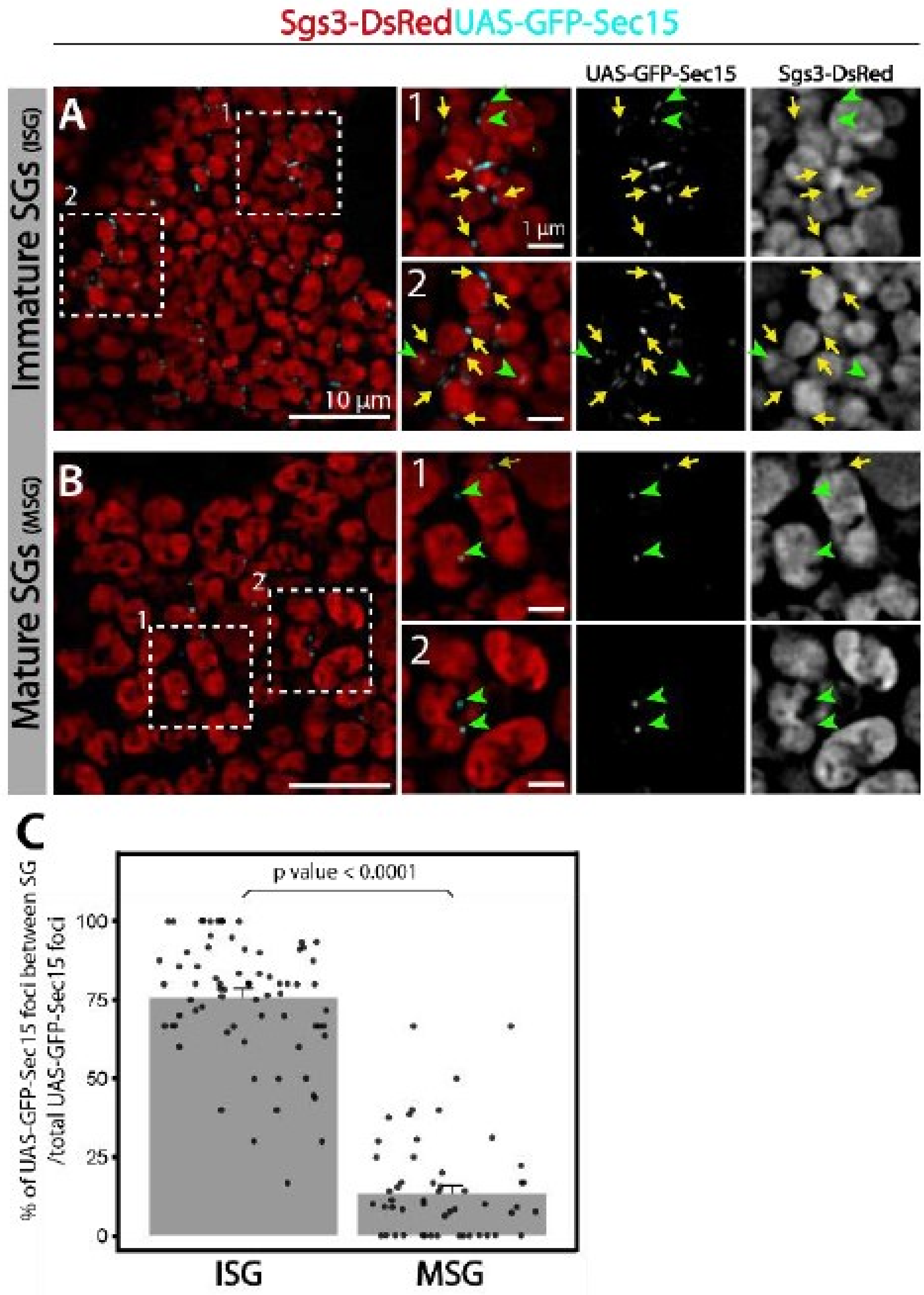
Author response image 2.

Criteria utilized to define Sec15 focithat were“associated” or“not associated” withthe ERin the experiments of Figure 5A-Bof the manuscript.When the distance between maximal intensities of GFP-Sec15 and KDEL-RFP signals was equal or less than 0.6 □m, the signals were considered “associated”. When the distance was more than 0.6 □m, the signals were considered “not associated”.



Author response image 3.

(A) GFP-Sec15 foci (cyan) and SGs (red) are shown in cells bearing Immature SGs or (B) with mature SGs. Yellow arrows indicate GFP-Sec15 foci localized in between SGs; green arrowheads indicate GFP-Sec15 foci that arenot in between SGs. (C) Quantification of the percentage (%) of Sec15 foci localized in between SGs respect to the total number of Sec15 foci in cells filled with immature SGs (ISG)vs cells with mature SGs (MSG).



It is interesting to mention that previous evidence from mammalian cultured cells (Yeaman et al, 2001) show that the exocyst localizes both at the trans-Golgi network and at the plasma membrane, weighing in favour of our claim that the exocyst is required at various steps of the exocytic pathway. Thus, the exocyst may play multiple roles in the secretion pathway in other biological models as well. This concept has now been included at the Discussion section of the revised version of the manuscript (lines 359-368).

To make the conclusions of our work clearer, in the revised version of the manuscript, we have now included a graphical abstract, summarizing the dynamic localization of the exocyst in relation to the processes of SG biogenesis, maturation and exocytosis reported in our work.

> *(3) Instead, it is possible that defects in Golgi traffic and granule homotypic fusion are not due to direct involvement of the exocyst in these processes, but secondary to a defect*

> *in canonical exocyst roles at the plasma membrane. A block in the last step of glue exocytosis could perhaps propagate backward in the secretory pathway to disrupt Golgi complexes or cause poor cellular health due to loss of cell polarity or autophagy.*

We thank the reviewer for these thoughtful comments. We have performed a number of additional experiments to assess "cellular health" or to identify possible defects in cell polarity after knock-down of exocyst subunits. These new data have been included in new supplementary figures 5 and 6 of the revised version of the manuscript (please see below).

In our view, the precise localization of GFP-Sec15 at the Golgi complex (Figure 5C-E), as well as in between immature secretory granules (Figure 7B-D), argues in favour of a direct involvement of the exocyst in SG biogenesis and homofusion respectively.

We truly appreciate the comment of the reviewer raising the possibility that the defects that we observe at early steps of the pathway (SG biogenesis and SG maturation) may actually stem from a backward effect of the role of the exocyst in SG-plasma membrane tethering. We wish to respectfully point out that the processes of biogenesis, maturation and plasma membrane tethering/fusion of SGs do not occur simultaneously in the Drosophila larval salivary gland in vivo, as they do in other secretory model systems (i.e. cell culture). In this regard, the experimental model is unique in terms of synchronization. In each cell of the salivary gland, the three processes (biogenesis, maturation and exocytosis) occur sequentially, and controlled by developmental cues. At the developmental stage when SGs fuse with the plasma membrane, SG biogenesis has already ceased many hours earlier: SG biogenesis occurs at 96-100 hours after egg lay (AEL), SG maturation takes place at 100-112 hours AEL, and SG-plasma membrane fusion happens only when all SGs have undergone maturation and are ready to fuse with the plasma membrane at 116-120 h AEL. Thus, in our view it is not conceivable that a defect in SG-plasma membrane tethering/fusion (116-120 h AEL) may affect backwards the processes of SG biogenesis or SG maturation, which have occurred earlier in development (96-112 h AEL).

As suggested by the reviewer, we have analysed several markers of cellular health and cell polarity, comparing conditions of exocyst subunit silencing (exo70RNAi, sec3RNAi or exo84RNAi) with wild type controls (whiteRNAi). These new data are depicted in Supplementary Figures 5 and 6, and described in lines 172-179 of the Results section of the revised version of the manuscript. Noteworthy, for these experiments we have applied silencing conditions that block secretory granule maturation, bringing about mostly immature SGs. Our analyses included: 1) Subcellular distribution of PI(4,5)P2, 2) subcellular distribution of the tetraspanin CD63, 3) of Rab11, 4) of filamentous actin, and 5) of CD8. We have also compared 6) nuclear size and nuclear general morphology, 7) the number and distribution of mitochondria, 8) morphology and subcellular distribution of the cis- and 9) trans-Golgi networks. Finally, 10) we have compared basal autophagy in salivary cells with or without knocking down exocyst subunits. The markers that we have analysed behaved similarly to those of control salivary glands, suggesting that the observed defects in regulated exocytosis indeed reflect different roles of the exocyst in the secretory pathway, rather than poor cellular health or impaired cell polarity.

Our conclusions are in line with previous studies in which apico-basal polarity, Golgi complex morphology and distribution, as well as apical membrane trafficking were also evaluated in exocyst mutant backgrounds, finding no anomalies (Jafar-Nejad et al, 2005).

Conversely, in studies in which apical polarity was disturbed by interfering with Crumbs levels, SG biogenesis, maturation and exocytosis were not affected (Lattner et al, 2019), indicating that these processes not necessarily interfere with one another.

> *(4) Final recommendation: In the absence of stronger evidence for these other exocyst roles, I would suggest focusing the study on the canonical role (interesting, as it was*

> *previously reported that Drosophila exocyst had no function in the salivary gland and limited function elsewhere [DOI: 10.1034/j.1600-0854.2002.31206.x]), and leave the alternative roles for discussion and deeper study in the future.*

We appreciate the reviewerś recommendation. However, we believe that the major strength of our work is the discovery of non-canonical roles of the exocyst complex, unrelated to its function as a tethering complex for vesicle-plasma membrane fusion. We believe that in the new version of our manuscript, we provide stronger evidence supporting the two novel roles of the exocyst:

a) Its participation in maintaining the normal structure of the Golgi complex, and b) Its function in secretory granule maturation.

> ***Reviewer 2:***
>
> *(5) General comment: A key strength is the breadth of the assays and study of all 8 exocyst subunits in a powerful model system (fly larvae). Many of the assays are quantitated and roles of the exocyst in early phases of granule biogenesis have not been ascribed.*

We are grateful that the reviewer appreciates the novelty of our contribution.

> *(6) However there are several weaknesses, both in terms of experimental controls, concrete statements about the granules (better resolution), and making a clear conceptual framework. Namely, why do KD of different exocysts have different effects on presumed granule formation*

The reviewer has raised a point that is central to the interpretation of all our data throughout the manuscript. The short answer is that the extent of RNAi-dependent silencing of exocyst subunits determines the phenotype:

1. Maximum silencing affects Golgi complex morphology and prevents SG biogenesis. 2) Intermediate silencing blocks SG maturation, without affecting Golgi complex morphology and SG biogenesis. 3) Weak silencing blocks SG tethering and fusion with the plasma membrane, without affecting Golgi complex morphology, SG biogenesis or SG maturation.

In other words, 1) Low levels of exocyst subunits are sufficient for normal Golgi complex morphology and SG biogenesis. 2) Intermediate levels of exocyst subunits are sufficient for SG maturation (and also sufficient for SG biogenesis). 3) High levels of exocyst subunits are required for SG tethering and subsequent fusion with the plasma membrane.

Based on the above notion, we have exploited the fact that temperature can fine-tune the level of Gal4/UAS-dependent transcription, thereby achieving different levels of silencing, as shown by Norbert Perrimon et al in their seminal paper “the level of RNAi knockdown can also be altered by using Gal4 lines of various strengths, rearing flies at different temperatures, or via coexpression of UAS-Dicer2” (Perkins et al, 2015).

We found in our system that indeed, by applying appropriate silencing conditions (RNAi line and temperature) to any of the eight subunits of the exocyst, we have been able to obtain one of the three alternative phenotypes: Impaired SG biogenesis, or impaired SG maturation, or impaired SG tethering/fusion with the plasma membrane.

These concepts are summarized below in Author response image 4. Please see also at point 26, the general comment of Reviewer #3.

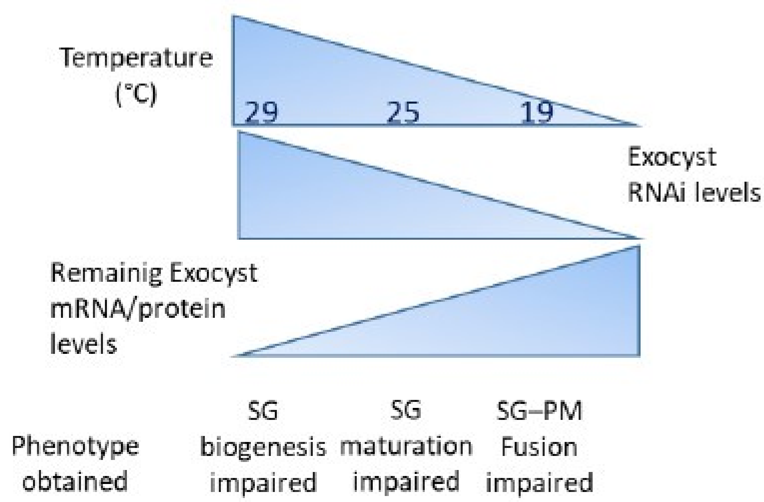
We have conducted qRT-PCR assays to provide experimental support to the notions summarized above in Author response image 4. We measured the remaining levels of mRNAs of some of the exocyst subunits, after inducing RNAi-mediated silencing at different temperatures, or with different RNAi transgenic lines. The remaining RNA levels after silencing correlate well with the observed phenotypes, following the predictions of Author response image 4 and summarized in Author response image 5. These new data are now shown in Supplementary Figure 2 of the revised version of the manuscript, and described in lines 153-159 at the Results section.

> *(7) Why does just overexpression of a single subunit (Sec15) induce granule fusion?*

The reviewer raises a very important point. Based on available data from the literature, Sec15 behaves as a seed for assembly of the holocomplex and it also mediates the recruitment of the holocomplex to SGs through its interaction with Rab11 (Escrevente et al, 2021; Bhuin and Roy, 2019; Wu et al, 2005; Zhang et al, 2004; Guo et al, 1999). Thus, overexpression of Sec15 is expected to enhance exocyst assembly, thereby potentiating the activities carried out by the complex in the cell, including SG homofusion. In the revised version of the manuscript we have also performed the overexpression of Sec8, finding that, unlike Sec15, Sec8 fails to induce homotypic fusion. These results were expected, as they confirm that Sec8 does not behave as a seed for mounting the whole complex. These new data have been included in Figure 7E-H, and are described in text lines 221-229 of the Results section.

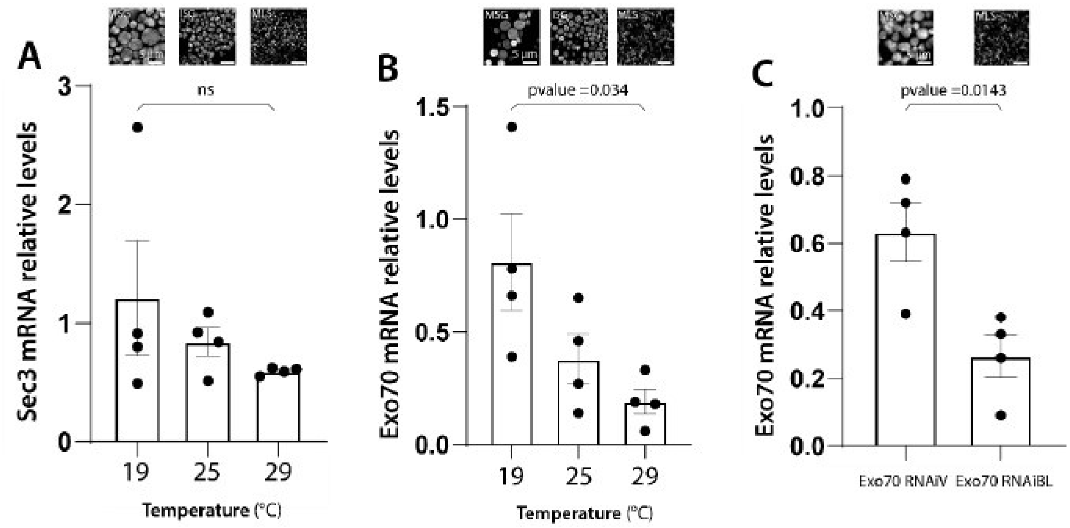
Author response image 4.

Conceptual model of RNAi expression at different temperatures , remaining levels of mRNA/protein levels and phenotypes obtained at each temperature.



Author response image 5.

qRT-PCR assays presented in Supplementary Figure 2 are shown in combination with the phenotypes observed at each of the conditions analyzed. Note the correlation between phenotypes and the extent of mRNA downregulation.



> *(8) While the paper is fascinating, the major comments need to be addressed to really be able to make better sense of this work, which at present is hard to disentangle direct vs. secondary effects, especially as much of the TGN seems to be altered in the KDs.*

We hope that our response to point 6) has helped to clarify this important point raised by the Reviewer. After applying silencing conditions where normal structure of the trans-Golgi network is impaired, SG biogenesis does not occur. Thus, since SGs do not form, it is not conceivable to detect defects in SG maturation or SG fusion with the plasma membrane in the same cell.

> *(9) The authors conveniently ascribe many of the results to the holocomplex, but their own data (Fig. 4 and Fig. 6) are at odds with this.*

This is another central point of our work, so we thank the reviewer for his/her comment. In Figures 4A, 7A and 9A of the revised version of the manuscript, we show that, by inducing appropriate levels of silencing of any of the 8 subunits of the exocyst, each of the three alternative phenotypic manifestations can occur. In our opinion, this argues in favour of a function for the whole exocyst complex in each of the three specific activities proposed in our study: 1) SG biogenesis, 2) SG maturation, and 3) SG tethering/fusion with the plasma membrane. In detailed characterizations of these three phenotypes performed throughout the study, we decided to induce silencing of just two or three of the subunits of the exocyst, assuming that the whole complex accounts the mechanisms involved.

> *Major comments*
>
> *(10) Resolution not sufficient. Identification of "mature secretory granules" (MSG) in Fig. 3 is based on low-resolution images in which the MSG are not clearly seen (see control in Fig. 3A) and rather appear as a diffuse haze, and not as clear granules. There may be granules here, but as shown it is not clear. Thus it would be helpful to acquire images at higher resolution (at the diffraction limit, or higher) to see and count the MSG.*

We thank the reviewer for raising this point, as it may not be straightforward to the reader to identify the SGs throughout the figures of our study. To make it clearer, in Figure 3A (magnified insets on the right), we have delimitated individual SGs with a green dotted line, and included diagrams (far right), which we hope will help the identification of SGs. In Figure 3B, we show that after silencing Sec84, a mosaic phenotype was observed: In some cells SGs fail to undergo maturation, and remain smaller than normal. In other cells of this mosaic phenotype, biogenesis of SGs was impaired and the fluorescent cargo remained trapped in a mesh-like structure (that we later show that corresponds to the ER). The dotted line marks individual SGs, and the diagrams included on the right intend to help the interpretation of the phenotype. The mesh-like structures where Sgs3-GFP was retained are also marked with dotted line, and schematized on the right. These new schemes are described in the Figure 3 caption of the revised version of the manuscript.

We wish to mention that all the confocal images depicted in this figure and throughout the manuscript have been captured at high resolution, with a theoretical resolution limit of 168177nm (d = γ/2NA). Given that secretory granules range from 0.8-7μm in diameter, the resolution is more than sufficient to clearly resolve these structures.

> *(11) Note: the authors are not clear on which objective was used. Maybe the air objective as the resolution appears poor).*

In this particular figure, we have utilized a Plan-Apochromat 63X/1.4NA oil objective of the inverted Carl Zeiss LSM 880 confocal microscope (mentioned in materials and methods).

> *(12) They need to prove that the diffuse Sgs3-GFP haze is indeed due to MSG.*

If we interpret correctly the concern of the reviewer, what he/she calls "diffuse haze" is actually the distribution of Sgs3-GFP within individual SGs, which, as previously reported by other authors, is not homogeneous at this stage (Syed et al. 2022). We hope that the diagrams that we have included in Figure 3 A, B (point 10) will help the readers interpreting the images.

> *(13) Related it is unclear what are the granule structures that correspond to Immature secretory granules (ISG) and cells with mesh-like structures (MLS)?*

We are confident that the diagrams now included in Figure 3A and B will help the interpretation, and particularly to identify immature granules and the mesh-like structure generated after silencing of exocyst subunits.

> *(14) Similarly, Sgs3 images of KD of 8 exocyst subunits were interpreted to be identical, in Fig. 4, but the resolution is poor.*

We hope that the issue related to resolution of our images has been properly addressed in the response to point 10) of this letter. In Figure 4A, we show that after silencing of any of the 8 subunits (with the appropriate conditions), in all cases SG biogenesis was impaired, and Sgs3GFP was instead retained in a mesh-like structure. Images obtained after silencing different exocyst subunits are of course not identical, but in all cases, a mesh-like structure has replaced the formation of SGs (Figure 4A). Hopefully, the diagrams now included in Figure 3A and B help the correct interpretation of the phenotypes throughout the study.

To demonstrate that the structure in which Sgs3-GFP was retained upon exocyst complex knockdown corresponds to the ER, we performed a colocalization analysis between Sgs3-GFP and the ER markers GFP-KDEL or Bip-sfGFP-HDEL, after which we calculated the Pearsons Coefficient, which indicated substantial colocalization (Figure 4B-G and Supplementary Figures 7 and 8). These new data are described in lines 196-199 of the revised version of the manuscript. To facilitate the visualization of the results, in the revised version of the manuscript we have included magnified cropped areas of the images shown in Figure 4A.

> *(15) What is remarkable is a highly variable effect of different subunit KD on the percentage of cells with MLS (Fig. 4C). Controls = 100 %, Exo70=~75% (at 19 deg), Sec3 =*

> *~30%, Sec10 = 0%, Exo84 = 100% ... This is interesting for the functional exocyst is an octameric holocomples, thus why the huge subunit variability in the phenotypes? The trivial explanation is either: i) variable exocyst subunit KD (not shown) or ii) variability between experiments (no error bars are shown). Both should be addressed by quantification of the KD of different proteins and secondly by replicating the experiments.*

We agree with the reviewer statement. We believe that both, variability of KD efficiency (i) and variability between experiments (ii) contribute to the variable effect observed after knocking down the different subunits. As detailed in the response to point 6), we have performed qRT-PCR determinations to confirm that the severity of the phenotype depends on the efficiency of RNAimediated silencing. We chose to analyse in detail the effect on the subunits exo70 and sec3, which were those with the highest phenotypic differences between the three silencing temperatures utilized. We found that as expected, the levels of silencing were temperaturedependent, being higher at 29°C and lower at 19°C. These data were included in Supplementary Figure 2, and described lines 153-159 of the Results section and also summarized in Author response images 4 and 5 of this rebuttal letter.

We thank the reviewer for his/her comment on the replication of experiments and statistics. We failed to include detailed numerical information in the original submission, such as the number of replicas and standard deviations of the data depicted in Figure 3C and Supplementary Figure 1, so we apologize for this omission. In the revised version of the manuscript, we have included a table (Supplementary Table 3) in which all the raw data of Figure 3C and Supplementary Figure 1, including standard deviations, are now depicted.

> *(16) If their data holds up then the underlying mechanism here needs to be considered.*
>
> *(Note: there is some precedent from the autophagy field of differential exocyst effects)*

Our proposed mechanism is essentially that the holocomplex is required for multiple processes along the secretory pathway. Each of these actions (Golgi structure maintenance, SG maturation and SG tethering/fusion with the plasma membrane) requires different amounts of holocomplex activity, being this the reason why each phenotype manifests at different levels of RNAi-mediated silencing (Author response image 4 of this letter). The model predicts that Golgi structure maintenance requires minimal levels of complex activity, and that is why strong knock-down of exocyst subunits is required to obtain this phenotype. In line with our results, it has been reported that other tethering complexes of the CATCHR family are also required for maintaining Golgi cisternae stuck together (D'Souza et al, 2020; Khakurel and Lupashin, 2023; Liu et al, 2019). One possibility is that the exocyst may play a redundant role in the maintenance of the normal structure of the Golgi complex, along with other CATCHR complexes. This potential redundancy could explain why severe exocyst knock-down is required to observe structural anomalies at this organelle. On the other end of the spectrum, we propose that tethering/fusion with the plasma membrane is very susceptible to even slight reduction of complex activity, so that mild RNAi-mediated silencing is sufficient to provoke defects in this process. This proposed model is depicted in Author response image 4 and discussed in lines 395-405 of the Discussion section.

> *(17) In the salivary glands the authors state that the exocyst is needed for Sgs3-GFP exit from the ER. First, Pearson's coefficient should be shown so as to quantitate the degree of ER localizations of all KDs.*

We thank the reviewer for this comment that helped us to strengthen the observation that when SG biogenesis is impaired, Sgs3-GFP remains trapped in the ER. In the revised version of the manuscript, we have calculated Pearsonś coefficient to assess colocalization between ER markers (GFP-KDEL or Bip-sfGFP-HDEL) and Sgs3-GFP in salivary gland cells that express sec15RNAi. The Pearson's coefficient was around 0.6 for both ER markers, indicating that colocalization with Sgs3-GFP was substantial (Supplementary Figure 8, text lines 196-199 of the Results section).

> *(18) Second, there should be some rescue performed (if possible) to support specificity.*

As suggested by the reviewer, we have performed a rescue experiment of the phenotype provoked by the expression of sec15 RNAi, which consisted on the retention of Sgs3-GFP in the endoplasmic reticulum: Expression of Sec15-GFP reverted substantially the ER retention phenotype, rescuing SG biogenesis and also SG maturation in most cells (over 60% of the cells). These new data are now shown in Supplementary Figure 4, and described in lines 168-171 of the Results section.

> *(19) Third, importantly other proteins that should traffic to the PM need to be shown to traffic normally so as to rule out a non-specific effect.*

We have addressed this issue (also mentioned by Reviewer #1), by analyzing the localization of a number of polarization markers, finding that the overall polarization of the cell was not affected by loss of function of exocyst subunits. Please, see our response to the point 3) raised by Reviewer #1. The new data showing cell polarization markers are shown in Supplementary Figure 6 of the revised version of the manuscript, and described on text lines 172-179 of the Results section.

> *(20) It is unclear from their model (Fig. 5) why after exocyst KD of Sec15 the cis-Golgi is more preserved than the TGN, which appears as large vacuoles. This is not quantitated and not shown for the 8 subunits.*

We thank the reviewer for this relevant comment. We agree that the phenotype of either, sec15 or sec3 loss-of-function cells manifests differently with cis-Golgi and trans-Golgi markers. While the cis-Golgi marker looked fragmented and aggregated, the trans-Golgi marker adopted a swollen appearance. However, in our view, the different appearance of the two markers does not necessarily imply that one compartment is more preserved than the other. In the revised version of the manuscript, we have quantified the penetrance of the phenotypes provoked by sec15 or sec3 silencing, using both cis-Golgi and trans-Golgi markers. In both cases, the penetrance was high, although even higher with the trans-Golgi marker. These new data are now depicted in Supplementary Figure 9 of the revised version of the manuscript.

It is interesting to mention that in HeLa cells, as well as in the retinal epithelial cell line hTERT, Golgi phenotypes similar to those we have described here have been reported after loss-offunction of other tethering complexes, which were shown to maintain the Golgi cisternae stuck together, including the GOC and GARP complexes (D'Souza et al, 2020, Khakurel and Lupashin, 2023; Shijie Liu et al, 2019). As we did throughout our work, not every aspect of the analysis included the silencing of all eight subunits. In this case, we chose to silence Sec3 and Sec15. Please note that we have modified the model depicted in Figure 6E-F, to highlight the cis- and transGolgi phenotypes upon exocyst knock-down, as well as the localization of the exocyst in cisternae of the Golgi complex.

> *(21) Acute/Chronic control: It would be nice to acutely block the exocyst so as to better distinguish if the effects observed are primary or secondary effects (e.g. on a recycling pathway).*

We thank the reviewer for raising this important issue. To address this point, and to be able to induce silencing of exocyst subunits at specific time intervals of larval development, we utilized a strategy based on a thermosensitive variant of the Gal4 inhibitor Gal80 (Gal80ts) (Lee and Luo, 1999). We blocked Gal4 activity (and therefore RNAi expression) by maintaining the larvae at 18 °C during the 1st and 2nd instars (until 120 hours after egg lay), and then induced the activity of Gal4 specifically at the 3rd larval instar by raising the temperature to 29 ºC, a condition in which Gal80ts becomes inactive. After silencing the expression of sec3 or sec15 at the 3rd larval instar only, the phenotype was very similar to that observed after chronic silencing of exocyst subunits (larvae maintained at 29 ºC all throughout development, where Gal4 was never inhibited). These observations suggest that the defects observed in the secretory pathway after knock down of exocyst subunits reflect genuine functions of the exocyst in this pathway, rather than a secondary effect derived from impaired development of the salivary glands at early larval stages. These new results are now shown in Supplementary Figure 3, and described in manuscript lines 160-171 of the Results section.

> *(22) Granule homotypic fusion. Strangely over-expression of just one subunit, Sec15-GFP, made giant secretory granules (SG) that were over 8 microns big! Why is that, especially if normally the exocyst is normally a holocomplex. Was this an effect that was specific to Sec15 or all exocyst subunits? Is the Sec15 level rate limiting in these cells? It may be that a subcomplex of Sec15/10 plays earlier roles, but in any case this needs to be addressed across all (or many) of the exocyst subcomplex members.*

Please, see our response to point 7) of this letter. Sec15 is believed to act as a seed for the formation of the whole complex.

> *(23) In summary, there are clearly striking effects on secretory granule biogenesis by dysfunction of the exocyst, however right now it is hard to disentangle effects on ERGolgi traffic, loss of the TGN, and a problem in maturation or fusion of granules.*

As discussed in detail in our response to the point 3 raised by Reviewer #1, the secretory pathway is highly synchronized in each of the cells of the Drosophila salivary gland. SG biogenesis, SG maturation and SG fusion with the plasma membrane never occur simultaneously in the same cell. Thus, in a cell in which ER-Golgi traffic is impaired (and SG biogenesis does not occur), SGs do not exist, and therefore, they cannot exhibit defects in the process of maturation or fusion with the plasma membrane. In summary, we believe that our work has shown that in Drosophila larval salivary glands the exocyst holocomplex is required for (at least) three functions along the secretory pathway: 1) To maintain the appropriate Golgi complex architecture, thus enabling ERGolgi transport; 2) For secretory granule maturation: both, homotypic fusion and acquisition of maturation factors; 3) For secretory granule exocytosis: secretory granule tethering to enable subsequent fusion with the plasma membrane. As mentioned above (point 6 of this letter), these three functions require different amounts of the holocomplex, and therefore can be revealed by inducing different levels of silencing.

> *(24) It is also confusing if the entire exocyst holocomplex or subcomplex plays a key role*

The fact that, by silencing any of the subunits (with the appropriate conditions) it is possible obtain any of the 3 phenotypes (impaired SG biogenesis, impaired SG maturation or impaired SG fusion with the plasma membrane) argues in favour of a function of the complex as a whole in each of these three functions.

> ***Reviewer 3:***
>
> *(25) General comment: Freire and co-authors examine the role of the exocyst complex during the formation and secretion of mucins from secretory granules in the larval*

> *salivary gland of Drosophila melanogaster. Using transgenic lines with a tagged Sgs3 mucin the authors KD expression of exocyst subunit members and observe a defect in secretory granules with a heterogeneity of phenotypes. By carefully controlling RNAi expression using a Gal4-based system the authors can KD exocyst subunit expression to varying degrees. The authors find that the stronger the inhibition of expression of exocyst the earlier in the secretory pathway the defect. The manuscript is well written, the model system is physiological, and the techniques are innovative.*

We appreciate the reviewerś assessment of our work.

> *(26) My major concern is that the evidence underlying the fundamental claim of the manuscript that "the exocyst complex participates" in multiple secretory processes lacks direct evidence.*

We thank the reviewer for raising this important issue. We believe that the analysis of Sec15 subcellular localization during salivary gland development (Figures 5, 7B-D and 9E-F), in combination with the detailed analysis of the phenotypes provoked by loss-of-function of each of the exocyst subunits, provide evidence supporting multiple functions of the exocyst in the secretory pathway. We have also included 3D reconstructions and videos of GFP-Sec15 colocalization with Golgi and SG markers to support exocyst localization associated to these structures (Supplementary Videos 1-7), text lines 200-210; 216-221 and 303-305.

> *(27) It is clear from multiple lines of evidence, which are discussed by the authors, that exocyst is essential for an array of exocytic events. The fundamental concern is that loss of homeostasis on the plasma membrane proteome and lipidome might have severe pleiotropic effects on the cell.*

We agree with the reviewer that this is an important point that needed to be addressed. As discussed in detail above at the response to point 3 raised by Reviewer #1, we have analysed several plasma membrane markers (including a PI(4,5)P2 lipid reporter), and found that overall, plasma membrane integrity and polarity were not substantially affected (Supplementary Figure 6). In addition, we have analyzed several markers of general cellular “health” that indicate that salivary gland cells do not seem to be distressed by the reduction of exocyst complex activity (Supplementary Figure 5). These new data are described in lines 172-179 of the Results section.

> *(28) Perhaps the authors have more evidence that exocyst is important for homeotypic fusion of the SGs, as supported by the localisation of Sec15 on the fusion sites.*

We believe that the fact that, by silencing any of the exocyst subunits (with the appropriate conditions), immature smaller-than-normal granules were observed, argus in favour that the exocyst as a whole participates in SG homofusion (Figure 7A). In addition, we have included more images, quantifications, 3D reconstructions and videos of GFP-Sec15 localized just at the contact sites between immature SGs. We have quantified and compared GFP-Sec15 localization at immature SG vs its localization at mature SGs, finding that localizes preferentially at immature SGs, supporting a role of the exocyst as a tethering complex during homotypic fusion (shown Figure 7B-C and Supplementary Videos 4-6, and described in lines 216-221 of the Results section). Please see also our response to the point 2 raised by reviewer 1 in this rebuttal letter, and to Author response image 3 above in this letter.

> *(29) The second question that I think is important to address is, what exactly do the varying RNAi levels correspond to in terms of experiments, and have these been validated? Due to the fundamental claim being that the severity of the phenotype being correlated with the level of KD, I think validation of this model is absolutely essential.*

We thank the Reviewer for raising this important point, and agree it was lacking in the original version of our manuscript. As discussed in our response to the point 6) raised by Reviewer #2, we have performed qRT-PCR determinations for exo70 and sec3 mRNA levels after inducing silencing of these subunits at different temperatures, or with different RNAi transgenic lines. The remnant mRNA levels correlate well with the observed phenotypes. Please see Supplementary Figure 2 of the revised manuscript, and Author response image 5 of this rebuttal letter; described in lines 155-159 of the Results section.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *- The authors assert in the discussion that exocyst involvement in constitutive secretion is well documented. This is based on a very recent study in mammalian culture cells. Therefore, I would not dismiss the issue as completely settled. Furthermore, a previous study of Drosophila sec10 reported no roles outside the ring gland (DOI: 10.1034/j.1600-0854.2002.31206.x).*

We have included these observations in the Discussion section. Lines 326-329.

> *- A salivary gland screening by Julie Brill's lab reported exocyst components as hits (DOI: 10.1083/jcb.201808017).*

We have referred to this paper in the Discussion section. Lines 326-329.

> *- It should be explained in more detail what is measured in graphs 7C, F, and others quantifying fluorescence around secretory granules. Looking at the images, the decrease in Rab1 and Rab11 seems less convincing.*

We have made a clearer description of how fluorescence intensity was measured in the Methods section lines 558-561. Also, we have uploaded a source data file in which the raw data of each experiment used for quantifications are disclosed.

Please note that the data indicates that Rab11 levels are higher in sec5 (Figure 8J-L) and sec3 (supplementary Figure 11M-R).

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *No major issues.*
>
> *Writing - The authors should better frame their interpretations of other studies of the exocyst that include the role in autophagy, Palade body trafficking, and differential roles of the subunits.*

We have discussed these specific points in the Discussion section, lines 348-355 and 409-410.

> *Minor - Fig. 6A: Why are variable temperatures (19-29 deg C used for the 8 KD experiments)?*
>
> *Please show it all at the same temperature (control too).*

The need for the usage of specific temperatures to obtain specific phenotypes with each of the RNAi lines used was explained in point 6 of this letter.

Reviewer #3 (Recommendations For The Authors):

In the abstract, the authors refer to the exocytic process and go on to describe secretory granule biogenesis and exocytosis. However, there are many exocytic processes aside from secretory granule biogenesis, and I think the authors should clarify this.

Corrected in the Abstract. Lines 19-21

Page 17 Thomas, 2021 reference, there is a glitch with the reference.

Thanks for noticing. Fixed.

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